

Inflation Regime Channel of Stock–Bond Relationship: Cross-Country Evidence

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Abstract

We present a model linking stock and bond returns relationship to inflation regimes across major markets. A representative investor holds both asset classes, with returns driven by systematic inflation shocks and idiosyncratic noise. We show that stock-bond relationship strengthens, in absolute terms, in a high inflation regime, is strongly regime-dependent, and rises with the probability of transitioning into a high-inflation regime. Using this framework, we perform an extensive empirical exercise where we provide both time series and panel-based cross-country evidence to empirically examine the U.S., Eurozone, Japan, China, Switzerland, Canada, and the UK. For the time series, we find the strongest support for a positive and significant stock-bond relationship during high inflation in the US and Japan and some consistent support in the rest markets. The panel analysis gave the strongest evidence of this inflation regime-dependent stock-bond connection. Our empirical results show that the inflation-regime channel to explain the stock-bond connection has merits, holds across bond maturities, and provides formal inflation thresholds that have concrete relevance for asset allocation decisions.

Keywords: Stock–bond relation; Inflation regimes; Cross-country evidence.

JEL Classification: E31; E44; G11; G12; G15.

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1 Introduction

Episodes of high inflation have repeatedly undermined the diversification benefits of bonds, leaving investors exposed precisely when they rely on fixed income to hedge equity risk. During the 1970s and early 1980s when the global inflation was persistently high, stock-bond correlations were often positive, meaning that bonds provided little diversification benefit when inflation surged. Conversely, in the low-inflation environment of the 2000s and 2010s, correlations turned negative as falling yields tended to accompany equity downturns, restoring bonds' role as a hedge. The post-pandemic inflation spike of 2021-2022 vividly illustrated these regime effects: as inflation in the United States and Europe climbed to multi-decade highs, both stocks and government bonds suffered steep losses in tandem, breaking the traditional inverse relationship and delivering one of the worst 60/40 portfolio performances in nearly a century. This phenomenon is consistent with findings by Campbell and Ammer (1993), who show that inflation shocks can eliminate bonds' traditional hedging role against equity risk, especially when inflation becomes a priced factor for both asset classes. This breakdown of stock-bond diversification has highlighted the need to understand how inflation regimes reshape the stock-bond correlation and refocused attention on the conditions under which the two asset classes move in tandem rather than offsetting each other.

The negative correlation between stocks and bonds underpins classical asset allocation strategies. Yet, empirical evidence shows that this correlation is neither stable nor universal—it varies across time and macroeconomic conditions. In particular, inflation regimes and volatility significantly influence the interaction between equity and fixed income markets. Historical evidence by Bekaert and Engstrom (2010) further shows that during supply-driven inflation episodes, stock-bond correlations rise, undermining diversification benefits. However, most empirical studies to date have relied on linear specifications (Li, 2002) or imposed arbitrary thresholds to define inflation regimes (Yang, Zhou, & Wang, 2009), offering little structural insight into the mechanisms underlying this regime dependence. Furthermore, much of the evidence comes from single-country studies, leaving open the question of whether such dynamics generalize across economies.

This paper develops a continuous-time model to explain how inflation regimes affect the stock-bond correlation structure. We incorporate a regime-dependent, mean-reverting process for inflation and allow asset return sensitivities to inflation to differ by regime, and derive explicit expressions for correlation and its response to inflation risk. The model analytically demonstrates that stock-bond correlation is relatively low (even negative) when inflation remains near its low-regime mean, but it increases sharply once the economy transitions into a high-inflation regime. Moreover, the correlation rises with the perceived probability of

switching into high-inflation state, reflecting investor expectations of regime change. Our approach builds on the intuition of Burkhardt and Hasseltoft (2012), who suggest that the cyclical nature of inflation fundamentally affects stock–bond comovement, but here we formalize it into a structural, regime-aware model that yields testable predictions.

Our contribution is twofold. First, on the theoretical side, we formulate and prove propositions illustrating how inflation volatility, regime shifts, and investor expectations jointly shape the stock–bond relationship. Second, on the empirical side, we evaluate the model’s implications using data from multiple countries. We conduct regression analyses in seven major economies – including the United States, Japan, Canada, the Eurozone, the United Kingdom, Switzerland, and China – to test whether the stock-bond correlation behaves differently across inflation regimes. In doing so, we show that the inflation-regime channel generalizes across diverse macroeconomic environments, providing evidence of its relevance beyond any single country.

Methodologically, we employ a threshold regression approach to determine endogenously the inflation level that delineates low- and high-inflation regimes. In particular, we use Hansen’s (1999) panel threshold regression technique, which estimates a threshold value of inflation and allows the slope coefficients (including the stock–bond return linkage) to vary between regimes while accounting for unobserved heterogeneity via fixed effects. This approach lets the data reveal if and where a statistically significant regime break exists, rather than imposing a priori an arbitrary inflation cutoff. We apply the threshold model to both individual country time series and to a pooled panel of countries, leveraging a broad dataset that spans several decades and includes a range of bond maturities. Examining different maturity bonds – from short-term Treasury bills to long-term government bonds – also enables us to assess whether inflation regimes consistently affect stock–bond correlations across the yield curve.

Consistent with our theoretical predictions, the empirical results indicate that inflation regimes materially alter stock–bond correlations in most markets. In the United States and Japan, for example, we find clear evidence of regime-dependent correlation: when inflation is below the endogenously estimated threshold, stock and bond returns are negatively correlated or essentially uncorrelated, preserving the classic hedge property of bonds. Once inflation rises above the threshold, however, the stock-bond correlation turns positive in these countries – stocks and bonds begin moving in the same direction, undermining diversification. Canada, the Eurozone, and the United Kingdom also exhibit similar regime-dependent patterns, though the magnitude of the shift is smaller. Switzerland remains an exception, with little evidence of a regime effect, likely reflecting its stable inflation and safe-heaven status. For China, the results reveal additional complexity. Without controlling

for inflation, the stock-bond correlation appears weak and statistically insignificant. However, after controlling for inflation explicitly, the analysis identifies three distinct inflation thresholds, indicating that the stock–bond correlation exhibits nuanced, regime-dependent behavior across inflation intervals. This suggests that structural and policy factors in China shape the interaction between equities and bonds differently than in other economies.

Panel data results corroborate these patterns. For medium- and long-term maturities (5- and 10-year bonds), the panel threshold regressions reveal significant inflation thresholds: stock-bond correlations are negative in low-inflation periods but turn positive once inflation exceeds the threshold, underscoring the erosion of bonds’ hedging role. In contrast, the short-term (3-month) panel exhibits similar tendencies but with weaker and statistically insignificant regime effects, consistent with the muted inflation sensitivity of short-term treasuries.

This study provides a unified theoretical and empirical perspective on how inflation regimes govern the interplay between stock and bond returns. The evidence suggests that the widely presumed negative stock-bond correlation is in fact regime-contingent: it can swiftly reverse when inflation enters a higher regime, as recent experience has starkly demonstrated. By pinpointing the inflation thresholds at which the stock–bond relationship changes, our findings underscore that the diversification benefits of bonds are not static. Rather, they depend crucially on the prevailing inflationary regime – a consideration that is vital for informed asset allocation and risk management decisions in an era of volatile inflation.

2 Literature Review

The correlation between stock and bond returns plays a central role of portfolio theory and risk management. Classical portfolio strategies rest on the assumption of a stable, negative relationship between these two asset classes, which enables diversification (Campbell & Ammer, 1993). However, evidence over the past century and a half suggests that this relationship is not constant and does not apply to all. Historical data show that stock-bond correlations have varied substantially, often turning positive during periods of high inflation and monetary tightening and reverting to negative in times of low inflation and economic expansion (Yang et al., 2009; Li, 2002).

Research on stock–bond return dynamics and predictability has established that these relationships are influenced by macroeconomic conditions, investor preferences, and time-varying risk premia. Ang and Bekaert (2007) show that while stock returns exhibit some predictability, this is often unstable and regime-dependent, while Campbell and Thompson (2008) emphasize the fragility of out-of-sample forecasting models. Baele, Bekaert,

and Inghelbrecht (2010) demonstrate how macroeconomic risk factors—such as inflation, growth, and volatility—drive time-varying stock–bond correlations. Kojien, Lustig, and Van Nieuwerburgh (2017) examine both the cross-section and time series of stock and bond returns, underscoring the role of consumption-based risk factors. Connolly, Stivers, and Sun (2005) and Adrian, Crump, and Vogt (2019) further show that market stress and uncertainty contribute to nonlinear shifts in correlations and risk premia, especially during flight-to-safety episodes. Inflation expectations and real interest rates are also central to this literature. Barr and Campbell (1997) show how inflation risk affects UK nominal and index-linked bond yields, while Ang, Bekaert, and Wei (2008) explore the predictive role of expected inflation through the term structure of real rates. Relatedly, D’Amico, Kim, and Wei (2018) extract inflation compensation from TIPS, and Duffee (2018) decomposes Treasury yields into expected inflation, risk premiums, and term premiums. Theoretical contributions by Brennan and Xia (2002) and Campbell, Sunderam, and Viceira (2017) highlight how inflation uncertainty reshapes optimal asset allocation and the stock–bond risk-return tradeoff, although few studies estimate the thresholds that define high inflation regimes across financial markets.

Macroeconomic drivers and risk premia remain central to the joint behavior of stocks and bonds. Basu and Bundick (2017) incorporate uncertainty shocks into a New Keynesian framework, revealing asymmetric effects on asset prices, while Gilchrist and Zakrajšek (2012) link credit spreads to cyclical bond and equity risk. Campbell, Pflueger, and Viceira (2020) extend this macro-finance perspective by showing how growth, inflation, and monetary shocks jointly affect return dynamics—complementing Rudebusch and Swanson (2008, 2012), who use DSGE models to resolve bond premium puzzles. Consumption-based frameworks further clarify these dynamics: Campbell and Cochrane (1999) model habit formation, Epstein and Zin (1989) introduce recursive preferences, and Jermann (1998, 2010, 2013) adds production-based extensions, connecting shocks to term structures and equity premia. Constantinides (1990) and Lustig, Van Nieuwerburgh, and Verdelhan (2013) emphasize the roles of precautionary savings and time-varying consumption ratios in shaping discount rates. Investor sentiment and market fear have also been identified as drivers of comovement. Banerjee (2022) finds that COVID-19-related fear increased correlations, while Drechsler (2013), Chen, Michaux, and Roussanov (2020), and Brennan and Xia (2002) show that uncertainty shocks influence portfolio allocations and liquidity, altering stock–bond dynamics. Methodological innovations have improved analysis of these effects—Cox, Ingersoll, and Ross (1985) offer foundational equilibrium models, Cochrane and Piazzesi (2005) link bond risk premia to macro factors, Hansen and Sargent (2008) apply robust control under model uncertainty, and Van Binsbergen et al. (2012) extend recursive preferences in DSGE

contexts. Despite these advances, we are not aware of studies that employ a threshold regression framework—as we do in this paper—to explicitly estimate stock–bond correlation regimes based on inflation or other macroeconomic indicators. Our contribution is to identify such nonlinearities and regime shifts, especially under high-inflation conditions, offering new insight into asset comovement and portfolio construction.

Many mechanisms have been proposed to explain this variation. Inflation uncertainty has been identified as a key driver, increasing the joint sensitivity of stocks and bonds to macroeconomic shocks (Li, 2002). Scruggs and Glabadanidis (2003) argue that shifting risk premia, in response to changes in inflation uncertainty and equity market volatility, further contribute to this dynamic. Extending this view to a global context, d’Addona and Kind (2006) find that inflation uncertainty and real interest rate risk together help explain stock–bond correlations across countries.

The cyclical nature of inflation also plays an important role. When inflation is demand-driven and procyclical, stock–bond correlation tends to be negative, reflecting an economy in which growth supports stocks while higher rates weigh on bonds. Conversely, when inflation is supply-driven and countercyclical, both markets tend to perform poorly, producing a positive correlation (Bekaert & Engstrom, 2010; Burkhardt & Hasseltoft, 2012). Valckx (2004) adds that while news about future excess returns accounts for much of the volatility in both markets, inflation-related news remains a substantial contributor to their joint movements.

Macroeconomic regimes and policy reactions also shape the stock–bond relationship. Guidolin and Timmermann (2005) demonstrate that correlations differ markedly between bull and bear markets, supporting the notion of regime-switching behavior. Campbell, Sunderam, and Viceira (2014) highlight how changes in central banks’ responsiveness to inflation versus output contributed to the transition from positive to negative stock–bond correlations in the early 2000s. Similarly, Connolly, Stivers, and Sun (2005) emphasize the combined influence of equity market uncertainty and inflation in explaining time variation in correlations, underscoring the importance of investor expectations and policy response functions.

Market efficiency offers additional insights. Tolikas (2018) shows that stock returns tend to lead high-yield bond returns—but not investment-grade bonds—suggesting that equity markets adjust more quickly to information about future risks. During crises, heightened information asymmetry and risk aversion make both markets more sensitive to shared shocks, increasing their correlation. Short-lived “flight-to-safety” episodes, documented by Connolly et al. (2005) and Baur and Lucey (2009), temporarily reduce the correlation as investors rotate from equities into safer bonds. However, when crises persist and liquidity dries up, both asset classes may come under pressure, driving correlation higher (Baele, Bekaert, & Inghelbrecht, 2010). The COVID-19 pandemic further illustrated how systemic shocks amplify

stock-bond comovement, with emerging markets acting both as recipients and transmitters of global financial stress (Hu et al., 2024)

These dynamics have direct implications for portfolio construction. While traditional mean-variance optimization presumes a stable, negative correlation, Dopfel (2003) notes that this assumption often fails during episodes of elevated inflation, when both correlation and risk aversion shift. Ohmi and Okimoto (2016) observe that while correlations have trended downward in advanced economies over the long run, they are frequently interrupted by sharp, temporary spikes during inflationary or crisis episodes. Wu, Yeo, DiCiurcio, and Wang (2021) find that despite such transitory disruptions, the longer-run negative correlation tends to reassert itself, supporting the strategic role of bonds as diversifiers. Together, these findings suggest that both structural trends and short-term fluctuations must be accounted for in constructing resilient portfolios.

Beyond inflation and growth, other risk factors have been shown to influence stock-bond comovement. Koijen, Lustig, and Van Nieuwerburgh (2017) document that bond-market predictors of economic activity also explain the cross-section of stock returns, with value stocks particularly sensitive to business cycle shocks embedded in bond yields. This points to the business cycle as a priced state variable shaping the relationship. Bansal, Connolly, and Stivers (2014) find that spikes in equity risk, proxied by implied volatility, drive the correlation more negative, consistent with flight-to-safety behavior. Demirovic, Guermat, and Tucker (2017) further show that credit risk and equity volatility jointly determine the conditional correlation between stocks and corporate bonds, and that this link weakens under extreme uncertainty.

Global market structure and investor behavior have also played a role. Ohmi and Okimoto (2016) report that in 'safe haven' economies, correlations have trended downward, while in 'riskier' European economies, correlations spiked during the euro crisis, reflecting intensified flight-to-quality dynamics. Bolla (2017) demonstrates that the apparent outperformance of fundamental indexing strategies in global bond markets largely reflects higher exposure to risks such as duration, default, convexity, liquidity, and carry-trade, which themselves mirror structural imbalances in sovereign debt markets. Such imbalances—including concentrated sovereign risk and persistent debt accumulation—can erode bonds' hedging properties during crises. Similarly, Hu et al. (2024) show how systemic events such as the COVID-19 pandemic amplify stock-bond comovement via global financial linkages, with emerging markets both absorbing and transmitting shocks.

Theoretical models have sought to rationalize these patterns by linking stock and bond returns to macroeconomic risks and investor preferences (Campbell & Ammer, 1993). While such frameworks provide valuable intuition, they often abstract from regime-switching be-

havior and lack empirical validation outside the U.S. context. Recent empirical evidence suggests that stock-bond correlations are indeed regime-dependent (Yang et al., 2009; Baele & Van Holle, 2017), yet many studies rely on linear specifications or impose arbitrary thresholds to define regimes, limiting their explanatory power. This paper addresses these gaps by developing a continuous-time model that explicitly incorporates regime-dependent inflation volatility and transition probabilities, yielding closed-form predictions consistent with empirical patterns across a broader set of countries. In doing so, it contributes both a rigorous theoretical foundation and robust cross-country evidence on how inflation regimes shape stock-bond comovement, informing more resilient portfolio strategies under shifting macroeconomic conditions.

3 Model Framework

Consider a representative investor who purchases domestic assets and is assumed to have a portfolio without international exposure and with no exchange rate risk. Asset prices are expressed in local currency. The investor allocates wealth across a stock index (risky) and a nominal bond (risk-free in real terms with respect to default risk). The investor stock and bond held are assumed to be affected by shocks affected by market-wide inflation shocks and their respective idiosyncratic shocks, which are assumed to be uncorrelated with each other and with the inflation shocks.

Consider a representative domestic investor who allocates wealth between two financial assets: a broad-based equity index (stock) and a nominal government bond. All asset prices and returns are denominated in the domestic currency, and the investor is assumed to have no international exposure or exchange rate risk.

The nominal bond is considered free of default risk and backed by the sovereign government. Although labeled "risk-free" in credit terms, its return is not insulated from real economic risks—most notably, inflation. The investor perceives the bond as exposed to purchasing power risk through variation in inflation expectations and volatility.

Both the stock and the bond are assumed to be exposed to:

- **Systematic macroeconomic risk** via aggregate inflation shocks, which affect the economy-wide discount rate, firm cash flows, and bond pricing;
- **Idiosyncratic asset-specific shocks**, which capture firm-level, sectoral, or liquidity-related noise.

We assume that:

- Inflation shocks are modeled as a continuous-time stochastic process with regime-dependent volatility;
- The idiosyncratic shocks to stocks and bonds are modeled as orthogonal Brownian motions;
- All sources of randomness are mutually independent:

$$\text{Cov}(dW_t^S, dW_t^P) = \text{Cov}(d\pi_t, dW_t^S) = \text{Cov}(d\pi_t, dW_t^P) = 0.$$

This setup allows us to isolate the contribution of inflation—both its level and its volatility—to the comovement and risk properties of stock and bond returns. The investor is risk-averse and evaluates portfolio performance based on the statistical properties of these returns, particularly their correlation, volatility, and inflation sensitivity.

3.1 Inflation Dynamics

Let π_t denote the inflation rate, governed by a regime-dependent Ornstein–Uhlenbeck process. Inflation π_t follows a regime-dependent Ornstein–Uhlenbeck (OU) process given by:

$$d\pi_t = -\kappa_r(\pi_t - \bar{\pi}_r)dt + \sigma_{\pi,r}dW_t^\pi, \quad r \in \{L, H\} \quad (1)$$

where r denotes the inflation regime (Low or High), $\bar{\pi}_r$ is the regime-dependent mean, κ_r is the mean-reversion rate, that is, κ_r is the speed of mean reversion, and $\sigma_{\pi,r}$ is inflation volatility. The stationary variance for inflation is $\text{Var}(\pi_t) = \sigma_{\pi,r}^2/2\kappa_r$. Our choice for the OU process in the modelling framework is that it describes a process which becomes stationary over a long time, a description that characterizes inflation based on empirical evidence, see blah blah. where:

- $r \in \{L, H\}$ indexes the current inflation regime: low (L) or high (H),
- $\bar{\pi}_r$ is the regime-dependent long-run mean of inflation,
- $\kappa_r > 0$ governs the speed of mean reversion toward $\bar{\pi}_r$,
- $\sigma_{\pi,r} > 0$ is the regime-specific volatility of inflation shocks,
- W_t^π is a standard Brownian motion, independent of other sources of risk in the economy.

Stationary variance for inflation in regime r is:

$$\text{Var}(\pi_t^r) = \frac{\sigma_{\pi,r}^2}{2\kappa_r} \quad (2)$$

3.2 Asset Returns

Let dR_t^S and dR_t^B denote the instantaneous returns on stocks and bonds:

$$dR_t^S = \mu_S dt + \alpha_S^r d\pi_t + \sigma_S dW_t^S \quad (3)$$

$$dR_t^B = \mu_B dt + \alpha_B^r d\pi_t + \sigma_B dW_t^B \quad (4)$$

where α_S^r, α_B^r are regime-dependent inflation betas, and dW_t^S, dW_t^B, dW_t^π are standard Brownian motions with $\text{Cov}(dW_t^S, dW_t^B) = \text{Cov}(dW_t^S, dW_t^\pi) = 0$, and so on.

We assume $\alpha_B^r < 0$ in all regimes, reflecting the standard inflation sensitivity of nominal bonds, and allow α_S^r to have either sign, where $\alpha_S^r > 0$ means the generator of inflation is good for the stock market, i.e. demand driven inflation which stimulates the economy and risk assets, and $\alpha_S^r < 0$ means the type of inflation is bad for the stock market, i.e. supply driven inflation which depresses productivity and stifles risk assets.

Thus, our assumptions:

- $\text{Cov}(dW_t^S, dW_t^B) = 0$
- $\text{Cov}(dW_t^S, dW_t^\pi) = 0$, etc.
- $\alpha_B^r < 0$ for all r , α_S^r can vary in sign.

3.3 Correlation Expression

Instantaneous correlation in regime r is:

$$\rho_{SB}^r = \frac{\alpha_S^r \alpha_B^r \text{Var}(\pi_t)}{\sqrt{(\alpha_S^r)^2 \text{Var}(\pi_t) + \sigma_S^2} \cdot \sqrt{(\alpha_B^r)^2 \text{Var}(\pi_t) + \sigma_B^2}} \quad (5)$$

Our stylized model gives rise to several important and foundational propositions which we document below and explain the economic intuition behind them. For the proofs of these propositions, see the Appendix.

Before we present the proposition, we state an important *Lemma*. It permits a one-to-one validation of our continuous-time, instantaneous-moment model with lower-frequency, real-world data (e.g., monthly returns). Specifically, it shows that under geometric Brownian

motion (GBM) the correlation of instantaneous log-returns is preserved when we move to cumulative arithmetic returns over any fixed horizon.

3.4 Invariance of Correlation Under Different Return Frequencies

Lemma Let stock $S(t)$ and bond $B(t)$ follow correlated geometric Brownian motions with

$$\frac{dS(t)}{S(t)} = dR_t^S, \quad \frac{dB(t)}{B(t)} = dR_t^B,$$

and instantaneous return correlation ρ . For any fixed interval $T > 0$, define cumulative returns

$$R^S(T) = \int_0^T \frac{dS(t)}{S(t)}, \quad R^B(T) = \int_0^T \frac{dB(t)}{B(t)}.$$

Then

$$\text{Corr}[R^S(T), R^B(T)] = \rho.$$

Consequently, the correlation estimated from discrete-time returns over horizon T is a consistent estimator of the instantaneous correlation under the GBM assumption. In GBM, the instantaneous and cumulative return correlations coincide; therefore, the sample correlations calculated from the discrete data consistently estimate instantaneous correlation parameter ρ . This implies that under time-stable moments and correlations, insights about the model could be checked with data observed at different frequencies.

4 Propositions

Proposition 1: Inflation Regime-Dependent Stock-Bond Correlation

The comovement between stock and bond returns is regime-dependent and exhibits amplification in the high-inflation regime. Specifically, the absolute magnitude of the stock-bond correlation is stronger in the high-inflation regime than in the low-inflation regime for all admissible values of correlation.

Let ρ_{SB}^H and ρ_{SB}^L denote the instantaneous stock-bond correlation under high and low inflation regimes, respectively. Then:

- (i) In the positive correlation domain, where $\rho_{SB}^H, \rho_{SB}^L \in (0, 1]$, we have:

$$\rho_{SB}^H > \rho_{SB}^L$$

(ii) In the negative correlation domain, where $\rho_{SB}^H, \rho_{SB}^L \in [-1, 0)$, we have:

$$\rho_{SB}^H < \rho_{SB}^L \quad \text{and hence} \quad |\rho_{SB}^H| > |\rho_{SB}^L|$$

(iii) More generally, in absolute terms:

$$|\rho_{SB}^H| > |\rho_{SB}^L| \quad \text{for all } \rho_{SB}^H, \rho_{SB}^L \in [-1, 1]$$

The instantaneous correlation between stock and bond returns differs across inflation regimes and stronger in high inflation regimes

Economic Intuition. The proposition reflects a core mechanism in our model: when inflation is more volatile, it becomes the dominant driver of variation in both stock and bond returns. In high-inflation regimes, firms face greater uncertainty over future costs, pricing power, and discount rates, increasing the sensitivity of stock returns to inflation shocks. At the same time, nominal bonds—whose cash flows are fixed in nominal terms—experience larger price swings as markets react to changes in expected inflation and real interest rates.

Because both asset classes increasingly rely on the same source of risk—inflation—their returns begin to move more closely together. In statistical terms, the shared systematic component grows relative to idiosyncratic noise, raising the correlation coefficient. In contrast, in low-inflation regimes where inflation is stable and predictable, the inflation channel is muted, and asset-specific factors dominate return dynamics, resulting in lower correlation.

Proposition 2: Inflation Volatility Raises Stock-Bond Correlation

The correlation ρ_{SB} increases in the variance of inflation:

$$\frac{\partial \rho_{SB}}{\partial \sigma_\pi^2} > 0$$

Economic Intuition. This proposition captures the idea that the stock–bond correlation is governed by the relative importance of shared macroeconomic risk. When inflation volatility rises, the component of both stock and bond returns driven by inflation shocks becomes more prominent relative to their idiosyncratic sources of risk. Specifically, both asset classes respond to the same stochastic driver—unexpected inflation—and this common exposure becomes stronger as inflation fluctuations intensify.

Mechanically, the correlation increases because the numerator of the correlation formula—reflecting co-movement through inflation—rises quadratically with inflation volatility,

while the denominator (which includes idiosyncratic variances) increases more slowly. The result is a higher correlation coefficient.

Economically, in high-volatility environments, investors revise expectations about inflation and policy more frequently, causing joint repricing of both equities (through cash flow discounting and risk premia) and bonds (through expected real returns). Thus, higher inflation volatility strengthens the linkage between equity and bond markets through a shared macro risk channel.

Proposition 3: Regime Transition Probability Amplifies Correlation

Let p_H be the probability of being in high inflation. Then:

$$\frac{\partial \mathbb{E}[\rho_{SB}]}{\partial p_H} > 0$$

Let ρ_{SB}^L and ρ_{SB}^H denote the stock-bond return correlations in the low- and high-inflation regimes, respectively, and let $p_H \in [0, 1]$ be the probability (or long-run weight) of the high-inflation regime. Then the expected correlation across regimes is given by:

$$\mathbb{E}[\rho_{SB}] = (1 - p_H)\rho_{SB}^L + p_H\rho_{SB}^H.$$

Suppose that the magnitude of the correlation is strictly larger in the high-inflation regime, i.e., $|\rho_{SB}^H| > |\rho_{SB}^L|$. Then:

$$\frac{\partial \mathbb{E}[\rho_{SB}]}{\partial p_H} = \rho_{SB}^H - \rho_{SB}^L.$$

Therefore:

- If $\rho_{SB}^H > \rho_{SB}^L$, then $\frac{\partial \mathbb{E}[\rho_{SB}]}{\partial p_H} > 0$.
- Regardless of the sign, the magnitude $|\mathbb{E}[\rho_{SB}]|$ increases with p_H .

Economic Intuition. This proposition reflects the idea that the average stock–bond correlation is not only shaped by current inflation conditions but also by the investor’s expectations about future regime persistence. As the probability of being in the high-inflation regime p_H increases—either due to greater duration of the regime or a higher chance of transitioning into it—the expected influence of high-volatility inflation dynamics grows.

Since high-inflation regimes are associated with greater sensitivity of both stocks and bonds to inflation shocks, a higher weight on that regime shifts the average covariance

upward. At the same time, the conditional variances of stock and bond returns also rise, but the relative increase in the common inflation-driven component dominates, leading to a rise in expected correlation.

Economically, this means that when agents assign a greater likelihood to prolonged inflation stress—such as during periods of policy uncertainty, commodity shocks, or de-anchored inflation expectations—the markets increasingly price both asset classes in relation to inflation risk. This results in tighter co-movement between stock and bond returns, even if the current regime is not high-inflation. The investor’s forward-looking belief about regime transitions thus amplifies the expected return correlation across asset classes.

Proposition 4: Convex Amplification via Sensitivity Updates

If the inflation sensitivities α_S^H and α_B^H increase in the high-inflation regime probability p_H , then the expected correlation is convex in p_H :

$$\frac{\partial^2 \mathbb{E}[\rho_{SB}]}{\partial p_H^2} > 0$$

Economic Intuition. This proposition captures a nonlinear amplification effect arising from the investor’s belief updating under regime uncertainty. As the probability of being in a high-inflation regime p_H increases, not only does the weight on the high-correlation regime ρ_{SB}^H rise linearly, but if asset sensitivities to inflation α_S^H and α_B^H also increase with p_H —as investors adjust risk exposures, pricing models, or inflation-hedging strategies—then the rise in expected correlation accelerates.

The economic logic is that when inflation risk becomes more dominant in the investor’s belief system, market participants increasingly price assets through that shared macro risk channel. But crucially, their inflation betas are not fixed; they respond to perceived inflation persistence. For example, in environments where inflation appears structurally embedded, firms adjust pricing power assumptions, investors demand inflation protection, and both equity and bond markets become more tightly coupled to inflation news.

This endogenous feedback causes the effect of rising p_H to be more than proportional, resulting in a **convex relationship: small increases in inflation regime probability initially cause modest rises in expected correlation, but larger shifts in p_H trigger disproportionately stronger co-movement between stocks and bonds.

Proposition 5: Asymmetric Betas Increase Regime Gap

Let $\Delta\alpha = |\alpha_S^H - \alpha_S^L|$. Then:

$$\frac{\partial(\rho_{SB}^H - \rho_{SB}^L)}{\partial\Delta\alpha} > 0$$

Economic Intuition. This proposition formalizes the idea that when assets respond asymmetrically to inflation across regimes, the divergence in their comovement also grows. The absolute difference in stock inflation betas, $\Delta\alpha = |\alpha_S^H - \alpha_S^L|$, captures how sensitive equity returns are to inflation shocks in high- versus low-inflation environments. When this asymmetry increases, the structure of return comovement across regimes becomes more distinct, resulting in a larger gap between the high-regime and low-regime stock–bond correlations.

Economically, this reflects the fact that in low-inflation regimes, inflation may be only weakly priced into equity markets, while in high-inflation regimes, inflation becomes a dominant risk factor — directly affecting expected cash flows, discount rates, and risk premia. If bond sensitivities remain relatively stable while stock sensitivities rise sharply with regime shifts, the comovement structure becomes increasingly regime-specific. The correlation in the high-inflation regime grows more rapidly than in the low-inflation regime, amplifying the regime gap in observed correlations.

This result helps explain why shifts between inflation regimes can cause abrupt changes in stock–bond correlation dynamics, particularly when equity markets exhibit nonlinear sensitivity to macroeconomic shocks.

These propositions have implications for the relationship between the stock and bond markets. For the formal proofs of these propositions that resulted from our stylized model, please see the Appendix.

In the next section, we develop an empirical strategy to investigate the relationship of stock and bond market returns for distinct inflation regimes.

5 Cross-Country Empirical Evidence

The theoretical propositions developed in the previous section have important implications for the dynamic relationship between stock and bond markets under different inflation regimes. In this section, we present an empirical strategy designed to investigate how the relationship between stock and bond returns varies across inflation regimes.

We extend this analysis in a cross-country setting by examining both a country-by-country time series analysis and a panel analysis of major economies. Our empirical analysis uses the time series threshold and panel threshold regression techniques. In particular, we

explore whether regime-dependent inflation dynamics systematically affect the co-movement of stock and bond returns, consistent with the model’s predictions. To do so, we classify inflation regimes using a threshold-based approach and estimate time-varying correlations across regimes. This framework enables us to test whether the empirical behavior of asset correlations aligns with the theoretical expectation that high-inflation regimes amplify the comovement between equities and bonds.

5.1 Data Description and Summary Statistics

This section introduces the dataset used in the empirical analysis and presents summary statistics for the key variables. The dataset contains monthly observations of government bond returns at different maturities, inflation rates, and stock market returns, covering multiple countries and time periods.

Our empirical analysis focuses on some of the world’s largest capital markets—specifically the United States, Japan, Canada, the Eurozone, the United Kingdom, Switzerland, and China. Together, these markets account for an estimated \$105 trillion in total stock market capitalization, representing nearly 90% of the global equity market, which stands at about \$120 trillion according to MacroMicro. The combined size of their government bond markets is estimated at \$64 trillion, accounting for 80% of the global sovereign bond market, which is valued at approximately \$80 trillion. For our analysis, we consider the total return indices for both stock and bond markets. On the bond side, we include all available maturity segments for which data are accessible. All raw data come from Bloomberg covering a range of the asset markets across each country. The sample period differs across countries and asset classes, depending on data availability. Below, Table 5.1.1 presents the indices used for each economy, along with the maturity segmentation for bond markets:

Table 5.1.1: Stock and Bond Indices by Country and Sample Period

Country	Index Name	Type	Sample Period
US	S&P500 Index	Stock	1926–2021
US	USTBILL30 Index	Bond 30D	1926–2021
US	USTBOND5 Index	Bond 5Y	1926–2021
US	USTBOND20 Index	Bond 20Y	1926–2021
Japan	NKY Index	Stock	1970–2025
Japan	UISRMJ1E Index	Bond 10Y	1989–2025
Japan	UISRMJ3E Index	Bond 3M	1989–2023
Canada	SPTSX Index	Stock	1919–2025
Canada	SPFCGBT Index	Bond	1998–2025
Canada	SPFCAGG Index	Bond Aggregate	1998–2025
Canada	SPFICT3T Index	Bond 3M	2007–2025
Canada	SPFICTBT Index	Bond T-Bill	2007–2025
Canada	SPFCG15T Index	Bond 1–5Y	1998–2025
Eurozone	SX5E Index	Stock	1997–2025
Eurozone	EMTT3UC Index	Bond 0–3M	2003–2025
Eurozone	EXEA5 Index	Bond 1–3Y	2007–2025
Eurozone	EXEB5 Index	Bond 3–5Y	2007–2025
Eurozone	EXEC5 Index	Bond 5–7Y	2004–2025
Eurozone	EXED5 Index	Bond 7–10Y	2007–2025
Eurozone	EXEE5 Index	Bond 10–15Y	2007–2025
UK	UKX Index	Stock	1950–2025
UK	FTRFBGL Index	Bond 0–5Y	1990–2025
UK	FTRFBGM Index	Bond 5–15Y	2004–2025
UK	FTRFBGM1 Index	Bond 5–10Y	2004–2025
UK	FTRFBGM2 Index	Bond 10–15Y	1999–2025
UK	FTRFBGH Index	Bond 15Y+	1999–2025
Switzerland	SMI Index	Stock	1988–2025
Switzerland	SPFISWUT Index	Bond	1922–2025
China	SHASHR Index	Stock	2004–2025
China	SPBCNGOT Index	Bond	2006–2025
China	SPBCN1BT Index	Bond 10Y	2005–2025

For each country, we report descriptive statistics and examine pairwise correlations based on the specific samples employed in the subsequent regressions. Because the available time spans of the data differ across countries — and in some cases even within a single country, depending on the maturity and series — we compute descriptive statistics separately for each group of observations corresponding to the datasets used in the regressions. As a result, some countries, such as Canada and the Eurozone, appear with more than one group of summary statistics in this section to reflect the different subsamples analyzed later.

Table 5.1.2: Summary Statistics—U.S.

Variable	Obs	Mean%	Std. Dev.%	Min%	Max%
Treasury bond returns (20Y)	1150	0.50	2.50	-11.20	15.21
Treasury bond returns (5Y)	1150	0.40	1.20	-6.40	11.99
Treasury bond returns (30D)	1150	0.30	0.30	-0.10	1.29
Stock market	1150	0.70	5.30	-29.91	39.12
Inflation	1150	2.96	3.98	-10.69	19.71

Table 5.1.2 reports the descriptive statistics for the key variables examined in the U.S.. Government bond returns across different maturities clearly exhibit a term structure of interest rate risk. The 20-year Treasury bond delivers an average monthly return of 0.5%, accompanied by a standard deviation of 2.5%, whereas the 5-year bond has a slightly lower mean return of 0.4% and reduced volatility at 1.2%. In contrast, the 30-day Treasury bill is remarkably stable, with both its mean and standard deviation at just 0.3%, reflecting the minimal risk of short-term instruments. These patterns align with the expectation that longer-term bonds are more sensitive to interest rate fluctuations, while short-term securities are largely stabilized by monetary policy. The inflation rate demonstrates considerable variation, averaging 2.96% with a high standard deviation of 3.98%, indicating the presence of both inflationary and deflationary periods in the sample. Meanwhile, stock market returns display the highest volatility among the variables, with a standard deviation of 5.3% and a wide range from -29.9% to 39.1%, consistent with the inherently higher risk and return profile of equities compared to fixed-income assets.

Table 5.1.3: Correlation Matrix—U.S.

	20Y	5Y	30D	Stock market	Inflation
T-Bond returns (20Y)	1.0000				
T-Bond returns (5Y)	0.8191	1.0000			
T-Bond returns (30D)	0.0832	0.2054	1.0000		
Stock market	0.0655	0.0543	-0.0205	1.0000	
Inflation	-0.0054	0.0449	0.4263	-0.0338	1.0000

Table 5.1.3 reports the correlation matrix of the key variables in the U.S.. The 20-year and 5-year Treasury bond returns exhibit a strong positive correlation, whereas their correlations with stock market returns are positive but relatively weak. The 30-day Treasury

bill shows a slightly negative correlation with stock returns, indicating that short-term bonds behave differently from long-term bonds in their interaction with equity markets. Inflation is moderately correlated with short-term bond returns but shows little association with long-term bonds and stock returns, reflecting the distinct ways in which these assets respond to inflationary pressures. It is important to note that these simple correlations may obscure more complex relationships between the variables, as they do not fully capture the underlying dynamics. Our subsequent threshold analysis explores these hidden relationships in greater detail.

Table 5.1.4: Summary Statistics—Japan

Variable	Obs	Mean%	Std. Dev.%	Min%	Max%
UISRMJ1E return	391	0.23	1.11	-5.99	4.81
UISRMJ3E return	391	0.01	0.12	-0.54	0.67
Stock market	391	-0.07	6.05	-27.22	18.29
Inflation	391	0.47	1.23	-2.50	4.20

Table 5.1.4 presents the descriptive statistics for Japan’s main variables. The Japan 10Y Bond(UISRMJ1E) has an average monthly return of 0.23% with a standard deviation of 1.11%, reflecting moderate returns and relatively low volatility, typical of longer-term sovereign debt in a low-yield environment. In contrast, the Japan 3M Bond(UISRMJ3E) shows a much lower mean return (0.01%) and extremely low volatility (0.12%), consistent with the stability of short-term government securities. The stock market records a slightly negative average return (-0.07%) with substantially higher volatility (6.05%) and a wide range of returns (from -27.22% to 18.29%), highlighting the higher risk and return potential of equities. Inflation in Japan is subdued, averaging only 0.47%, with modest variability (1.23%), and remains within a relatively narrow range (from -2.50% to 4.20%), consistent with the country’s historically low-inflation environment. These figures illustrate the conservative risk–return trade-offs and muted inflation dynamics characteristic of the Japanese economy.

Table 5.1.5: Correlation Matrix—Japan

	UISRMJ1E	UISRMJ3E	Stock market	Inflation
UISRMJ1E return	1.0000			
UISRMJ3E return	0.5370	1.0000		
Stock market	-0.1116	0.1161	1.0000	
Inflation	0.0181	-0.0246	-0.0975	1.0000

Table 5.1.5 displays the correlation matrix for Japan’s main variables. The Japan 10Y Bond and Japan 3M Bond returns are moderately correlated (0.5370), reflecting their shared exposure to the sovereign debt market but also their differing sensitivities to interest rate movements. The correlations between bond returns and stock market returns are weak and mixed in sign: the 10Y bond return is slightly negatively correlated with stocks (-0.1116), while the 3M bond return is weakly positive (0.1161), suggesting that short-term bonds behave differently from long-term bonds in relation to equities. Inflation shows negligible correlations with both bonds (0.0181 with the 10Y bond and -0.0246 with the 3M bond) and a slightly negative correlation with stock market returns (-0.0975), consistent with the low and stable inflation environment in Japan. Overall, these results highlight the muted interactions among asset classes in the Japanese market and the differentiated behavior of short- and long-term bonds.

Table 5.1.6: Summary Statistics—Canada

Variable	Obs	Mean%	Std. Dev.%	Min%	Max%
Group A					
SPFICT3T return	199	0.12	0.11	-0.02	0.44
SPFICTBT return	199	0.12	0.12	-0.07	0.50
Stock market	199	0.20	4.06	-19.52	10.62
Inflation	199	2.17	1.60	-0.90	8.10
Group B					
SPFCGBT return	308	0.32	1.21	-3.35	4.73
SPFCAGG return	308	-0.03	1.13	-3.42	3.99
SPFCG15T return	308	0.27	0.59	-1.81	2.50
Stock market	308	0.35	4.35	-22.57	11.19
Inflation	308	2.15	1.39	-0.90	8.10

Table 5.1.6 presents the descriptive statistics of the main variables for Canada. Group A summarizes the results for the Canada Government 3M Treasury (SPFICT3T), Total Government Treasury Bill (SPFICTBT), stock market, and inflation. The Canada Government 3M Treasury return averages 0.12% per month with a standard deviation of 0.1%, indicating low returns with minimal volatility, which is consistent with the low-risk nature of short-term government debt. The Total Government Treasury Bill exhibits a similar mean return of 0.12% and slightly higher volatility at 0.12%, reinforcing the stability of short-term government instruments. The stock market return is substantially higher on average (0.20%) but also much more volatile, with a standard deviation of 4.06%, reflecting the higher risk associated with equities. Inflation averages 2.17% with moderate variability (1.60%), capturing periods of both price stability and fluctuations. Group B includes the Canada Government Bond (SPFCGBT), Canada Aggregate Bond (SPFCAGG), Canada Government 1-5Y Bond (SPFCG15T), stock market, and inflation. Among these, the Canada Government Bond exhibits the highest average return of 0.32% but also the largest volatility among bonds (1.21%), indicating the sensitivity of longer-term bonds to interest rate movements. The Canada Aggregate Bond shows a slightly negative mean return (-0.03%) with a standard deviation of 1.13%, suggesting its performance reflects a broad mix of bond instruments. The Canada Government 1-5Y Bond achieves a mean return of 0.27% and displays lower volatility (0.59%) compared to longer-term bonds, as expected. The stock market and inflation in Group B exhibit patterns similar to those in Group A, with stock returns averaging 0.35% and showing substantial variability (4.35%), while inflation remains stable at around 2.15%. These results highlight the expected trade-off between risk and return across asset classes and the term structure of interest rate risk in government bonds.

Table 5.1.7: Correlation Matrix—Canada

Group A					
	SPFICT3T	SPFICTBT	Stock market	Inflation	
SPFICT3T	1.0000				
SPFICTBT	0.9620	1.0000			
Stock market	-0.1841	-0.2596	1.0000		
Inflation	0.3277	0.2385	-0.1402	1.0000	

Group B					
	SPFCGBT	SPFCAGG	SPFCG15T	Stock market	Inflation
SPFCGBT	1.0000				
SPFCAGG	0.9902	1.0000			
SPFCG15T	0.8274	0.9272	1.0000		
Stock market	0.0793	0.1710	-0.0927	1.0000	
Inflation	-0.1137	-0.1200	-0.1068	-0.1048	1.0000

Table 5.1.7 displays the correlation matrix for the Canadian variables. In Group A, the returns on the Canada Government 3M Treasury and the Total Government Treasury Bill are highly positively correlated (0.9620), reflecting their shared exposure to short-term government securities. Both bonds exhibit weak and negative correlations with stock market returns (-0.1841 and -0.2596, respectively), suggesting that short-term government bonds may act as a hedge against equity risk. Inflation shows a moderate positive correlation with short-term bonds (0.3277 for the 3M Treasury and 0.2385 for the Treasury Bill), indicating some sensitivity of short-term yields to inflationary pressures. The stock market and inflation are weakly and negatively correlated (-0.1402), implying limited direct interaction. Group B focuses on the Canada Government Bond, Canada Aggregate Bond, and Canada Government 1-5Y Bond. As expected, these long-term and medium-term bonds are strongly correlated with each other, with coefficients exceeding 0.8, and the Government Bond and Aggregate Bond reaching almost perfect correlation (0.9902). Their correlations with stock returns are weak and mixed in sign (ranging from 0.0793 to -0.0927), suggesting that longer-term bonds do not systematically move with equities. Inflation shows mild negative correlations with the bonds in this Group (from -0.1137 to -0.1068), consistent with the notion that inflationary pressures tend to erode bond returns. Overall, these results highlight distinct patterns of comovement among asset classes and suggest that simple correlations may understate more

complex dynamics explored in subsequent analysis.

Table 5.1.8: Summary Statistics — Eurozone

Variable	Obs	Mean%	Std. Dev. %	Min%	Max%
Group A					
EMTT3UC return	258	0.08	0.13	-0.11	0.41
Stock market	258	0.30	4.87	-17.79	16.60
Inflation	258	2.13	1.96	-0.60	10.60
Group B					
EXEA5 return	216	0.10	0.45	-1.58	1.54
EXEB5 return	216	0.16	0.84	-3.08	2.73
EXEE5 return	216	0.12	2.17	-7.09	5.21
Stock market	216	0.11	5.16	-17.79	16.60
Inflation	216	2.13	2.14	-0.60	10.60
Group C					
EXEC5 return	245	0.22	1.17	-4.42	3.43
Stock market	245	0.26	4.95	-17.79	16.60
Inflation	245	2.14	2.01	-0.60	10.60
Group D					
EXED5 return	207	0.24	1.65	-5.89	4.57
Stock market	207	0.09	5.25	-17.79	16.60
Inflation	207	2.13	2.18	-0.60	10.60
Group E					
EMIEF5 return	252	0.25	3.25	-12.13	9.00
Stock market	252	0.24	4.91	-17.79	16.60
Inflation	252	2.14	1.98	-0.60	10.60

Table 5.1.8 reports the descriptive statistics for Eurozone government bonds, stock market, and inflation across five groups of bond maturities. In Group A, the Eurozone Government Bill 0-3M(EMTT3UC) exhibits a mean monthly return of 0.08% with very low volatility (0.13%), highlighting the extreme stability and low risk typical of short-term bills. The stock market in this Group shows a much higher mean return (0.30%) and substantial volatility (4.87%), while inflation averages 2.13% with moderate variation (1.96%). Group B examines the Eurozone Government Bonds 1-3Y(EXEA5), 3-5Y(EXEB5), and 10-15Y(EXEE5).

The 1–3Y bond has a mean return of 0.10% and low volatility (0.45%), while the 3–5Y bond achieves a higher return (0.16%) with increased risk (0.84%). The 10–15Y bond stands out with the highest return in this group (0.26%) and greater volatility (2.17%), reflecting higher duration risk. Stock market and inflation in this Group remain similar to Group A. In Group C, the Eurozone Government Bond 5–7Y(EXEC5) has an average return of 0.22% and volatility of 1.17%, showing a continued upward pattern in both return and risk compared to shorter maturities. Group D, which covers the Eurozone Government Bond 7–10Y(EXED5), further illustrates this trend, with a mean return of 0.24% and a higher standard deviation (1.65%), consistent with longer-term bonds being more sensitive to interest rate fluctuations. Finally, Group E, representing the Eurozone Government Bond 15Y and above(EMIEF5), exhibits the highest return (0.25%) and the largest volatility (3.25%) among all bond groups, underscoring the significant risk-return trade-off at long maturities. Across all Groups, the stock market consistently shows higher returns and substantially greater volatility than government bonds, while inflation remains stable at approximately 2.13–2.14%, with relatively small fluctuations.

Table 5.1.9: Correlation Matrix—Eurozone

Group A				
	EMTT3UCreturn	Stock market	Inflation	
EMTT3UCreturn	1.0000			
Stock market	-0.0444	1.0000		
Inflation	0.2122	-0.0531	1.0000	

Group B					
	EXEA5return	EXEB5return	EXEE5return	Stock market	Inflation
EXEA5return	1.0000				
EXEB5return	0.9384	1.0000			
EXEE5return	0.6962	0.8590	1.0000		
Stock market	0.0137	0.0746	0.2226	1.0000	
Inflation	-0.1686	-0.2101	-0.2406	-0.0536	1.0000

Group C			
	EXEC5return	Stock market	Inflation
EXEC5return	1.0000		
Stock market	0.0990	1.0000	
Inflation	-0.2172	-0.0529	1.0000

Group D			
	EXED5return	Stock market	Inflation
EXED5return	1.0000		
Stock market	0.1619	1.0000	
Inflation	-0.2351	-0.0533	1.0000

Group E			
	EMIEF5return	Stock market	Inflation
EMIEF5return	1.0000		
Stock market	0.1442	1.0000	
Inflation	-0.2370	-0.0522	1.0000

Table 5.1.9 presents the correlation matrices for Eurozone bonds, stock market, and inflation across five Groups. In Group A, the Eurozone Government Bill 0-3M shows weak correlations with both stock returns (-0.0444) and inflation (0.2122), reflecting its short-term, low-risk nature. Group B reveals very strong positive correlations among the bonds (up to 0.9384), while correlations with stocks remain weakly positive (maximum 0.2226) and correlations with inflation are moderately negative (as low as -0.2406), consistent with longer duration bonds' sensitivity to inflation. In Group C, the 5-7Y Government Bond is weakly positively correlated with stocks (0.0990) and moderately negatively correlated with inflation (-0.2172), similar to the pattern in Group B. Group D, representing the 7-10Y Government Bond, exhibits slightly higher positive correlation with stock returns (0.1619) and maintains a negative correlation with inflation (-0.2351), again showing the inflation risk of longer maturities. Finally, Group E, for the 15Y and above Government Bond, shows the highest positive correlation with stock returns (0.1442) and a strong negative correlation with inflation (-0.2370), underscoring the vulnerability of long-term bonds to inflationary pressures. Overall, bonds of similar maturities exhibit strong positive comovement, while their correlations with stocks remain weak and mixed, and their correlations with inflation are consistently negative, especially for longer-term bonds.

Table 5.1.10: Summary Statistics—UK

Variable	Obs	Mean%	Std. Dev. %	Min%	Max%
Group A					
FTRFBGLreturn	252	0.20	0.59	-3.17	2.46
FTRFBGMreturn	252	0.26	1.80	-9.83	5.24
Stock market	252	0.27	3.71	-14.86	11.65
Inflation	252	2.80	2.18	-0.10	11.10
Group B					
FTRFBGM1return	308	0.30	1.55	-9.38	4.73
FTRFBGM2return	308	0.31	2.18	-11.10	6.20
FTRFBGHreturn	308	0.25	3.33	-12.79	9.76
Stock market	308	0.10	3.91	-14.86	11.65
Inflation	308	2.50	2.08	-0.10	11.10

Table 5.1.10 summarizes the descriptive statistics for the UK. Group A reports the results for the UK Government Gilt Index 0-5Y(FTRFBGL), UK Government Gilt Index 5-15Y(FTRFBGL), stock market, and inflation. The short-term 0-5Y Gilt Index exhibits

an average return of 0.20% per month with low volatility at 0.59%, consistent with its lower exposure to interest rate fluctuations. In contrast, the medium-term 5-15Y Gilt Index has a higher average return (0.26%) and significantly greater volatility (1.80%), reflecting its longer duration and sensitivity to interest rate movements. The stock market yields an average return of 0.27%, but with much higher variability (3.71%), characteristic of equity markets. Inflation averages 2.80% with a standard deviation of 2.18%, indicating that the sample spans periods of both price stability and inflationary episodes. Group B focuses on the UK Government Gilt Index 5-10Y(FTRFBGM1), UK Government Gilt Index 10-15Y(FTRFBGM2return), and UK Government Gilt Index 15Y and above(FTRFBGH), alongside stock market and inflation. Among the gilt indices, the 5–10Y maturity exhibits the highest mean return (0.30%) and moderate volatility (1.55%), while the 15Y and above maturity has a slightly lower return (0.25%) but higher volatility (3.33%), consistent with the notion that longer-term bonds are more sensitive to interest rate risk. The stock market and inflation in Group B show similar patterns to Group A, with the stock market return averaging 0.10% and displaying notable volatility (3.91%), and inflation averaging 2.50% with moderate dispersion. These results highlight the expected trade-off between return and risk across maturities and asset classes in the UK market.

Table 5.1.11: Correlation Matrix—UK

Group A					
	FTRFBGLreturn	FTRFBGMreturn	Stock market	Inflation	
FTRFBGLreturn	1.0000				
FTRFBGMreturn	0.8699	1.0000			
Stock market	-0.0102	0.0151	1.0000		
Inflation	-0.0953	-0.1273	-0.0580	1.0000	

Group B					
	FTRFBGM1	FTRFBGM2	FTRFBGH	Stock market	Inflation
FTRFBGM1return	1.0000				
FTRFBGM2return	0.9748	1.0000			
FTRFBGHreturn	0.8623	0.9361	1.0000		
Stock market	-0.0419	-0.0251	0.0014	1.0000	
Inflation	-0.1268	-0.1365	-0.1874	-0.0175	1.0000

Table 5.1.11 reports the correlation matrix for the UK variables. In Group A, the UK Government Gilt Index 0-5Y and UK Government Gilt Index 5-15Y exhibit a strong positive correlation (0.8699), reflecting their overlapping exposure to medium-term sovereign debt. Their correlations with stock market returns are close to zero (-0.0102 and 0.0151, respectively), suggesting minimal interaction between government bonds and equities over this maturity spectrum. Inflation shows weak negative correlations with both gilt indices (-0.0953 and -0.1273) and a similarly modest negative relationship with stock returns (-0.0580), indicating limited contemporaneous co-movement among these variables. Group B focuses on longer maturities, namely the UK Government Gilt Index 5-10Y, 10-15Y, and 15Y and above. These gilt indices are highly correlated with each other, with coefficients exceeding 0.86, consistent with the similar risk exposures of medium- and long-term bonds. Correlations with stock returns remain negligible and slightly negative (ranging from -0.0419 to 0.0014), while inflation continues to display weak negative correlations with bond returns (from -0.1268 to -0.1874) and with the stock market (-0.0175). Overall, these results underscore the stability and co-movement within the sovereign bond market, as well as the limited direct association between fixed-income and equity returns in the UK.

Table 5.1.12: Summary Statistics—Switzerland

Variable	Obs	Mean%	Std. Dev.%	Min%	Max%
SPFISWUreturn	254	0.16	1.40	-4.80	5.76
Stock market	254	0.31	3.56	-12.03	9.64
Inflation	255	0.56	1.10	-1.40	3.50

Table 5.1.12 presents the descriptive statistics for Switzerland’s main variables. The Switzerland Government Bond(SPFISWU) delivers an average monthly return of 0.16% with a standard deviation of 1.40%, indicating modest returns with relatively low volatility, consistent with the safe-haven nature of Swiss sovereign debt. The stock market exhibits a higher mean return of 0.31%, but its volatility is more than double that of bonds, at 3.56%, reflecting the inherently greater risk associated with equities. Inflation remains subdued over the sample period, averaging 0.56% with low dispersion (1.10%) and a range from -1.40% to 3.50%, suggesting a relatively stable price environment in Switzerland. These patterns highlight the expected trade-off between risk and return across asset classes and underscore Switzerland’s reputation for monetary and financial stability.

Table 5.1.13: Correlation Matrix—Switzerland

	SPFISWUreturn	Stock market	Inflation
SPFISWUreturn	1.0000		
Stock market	-0.0162	1.0000	
Inflation	-0.0071	-0.1573	1.0000

Table 5.1.13 reports the correlation matrix of the key variables for Switzerland. The Switzerland Government Bond return exhibits a very weak and slightly negative correlation with stock market returns (-0.0162) and with inflation (-0.0071), suggesting that Swiss bonds move largely independently of equities and inflationary pressures. The correlation between stock returns and inflation is also negative and somewhat stronger (-0.1573), reflecting the tendency of equity markets to underperform in periods of rising prices. Overall, the low and negative correlations among these variables highlight the safe-haven properties of Swiss government bonds and the relatively muted interactions between asset classes in the Switzerland context.

Table 5.1.14: Summary Statistics—China

Variable	Obs	Mean%	Std. Dev.%	Min%	Max%
SPBCN1BTreturn	219	0.36	1.27	-3.74	6.30
SPBCNGOTreturn	219	0.33	0.64	-1.73	3.15
Stock market	219	0.10	7.27	-28.24	18.76
Inflation	219	2.30	1.98	-1.80	8.70

Table 5.1.14 presents the descriptive statistics for China's main variables. The China Government 10Y Bond(SPBCN1BT) exhibits an average monthly return of 0.36% with a standard deviation of 1.27%, reflecting modest returns and moderate volatility typical of longer-term sovereign debt. The China Government Bond(SPBCNGOT), which represents a broader mix of maturities, has a slightly lower mean return (0.33%) and lower volatility (0.64%), indicating its greater stability relative to the longer-term 10Y bond. The stock market shows a much lower average return of 0.10%, but with substantial volatility (7.27%) and a wide return range (from -28.24% to 18.76%), underscoring the higher risk inherent in equities. Inflation averages 2.30% with moderate variability (1.98%), ranging from -1.80% to 8.70%, consistent with episodes of both price stability and inflationary pressure during the sample period. These results highlight the differences in risk and return across asset classes in China's financial markets.

Table 5.1.15: Correlation Matrix—China

	SPBCN1BTreturn	SPBCNGOTreturn	Stock market	Inflation
SPBCN1BTreturn	1.0000			
SPBCNGOTreturn	0.6317	1.0000		
Stock market	-0.2674	-0.1236	1.0000	
Inflation	0.0435	-0.0007	-0.1933	1.0000

Table 5.1.15 presents the correlation matrix for the main variables in China. The China Government 10Y Bond and the China Government Bond returns are positively correlated (0.6317), reflecting their common exposure to sovereign debt markets, although the correlation is moderate rather than perfect, likely due to differences in maturity structure. Both bond returns exhibit negative correlations with stock market returns (-0.2674 and -0.1236, respectively), suggesting that Chinese government bonds tend to serve as a hedge against equity market risk, with the longer-term 10Y bond showing a stronger inverse relationship. Inflation is weakly correlated with both bonds (0.0435 for the 10Y bond and -0.0007 for the broader government bond), and its correlation with stock market returns is also negative and slightly stronger (-0.1933), indicating that inflationary pressures are associated with lower equity returns. These findings point to modest diversification benefits across asset classes and highlight the role of government bonds in stabilizing portfolios during equity market downturns.

In the panel data analysis, we construct three distinct panels, each using government bond returns of different maturities (3-month, 5-year and 10-year) as the dependent variable. The 3-month panel includes data from Japan, Canada, and the Eurozone over the period from 2007M03 to 2022M08. The 5-year panel draws on data for Canada, the Eurozone (3-5 years), and the United Kingdom (5-10 years), covering the period from 2007M01 to 2023M09. The 10-year panel comprises observations from Japan, China, the United Kingdom (5-15 years), and the Eurozone (7-10 years), spanning the period from 2008M01 to 2022M08. For all three panels, we additionally incorporate U.S. Bloomberg data, which does not distinguish between maturities. This choice reflects a data limitation: maturity-specific U.S. bond return data are available only for earlier periods and do not align with the time frame of our study. Consequently, we include the aggregate U.S. government bond return series to maintain consistency in the sample while acknowledging the lack of maturity differentiation in the U.S. data.

Table 5.1.16: Summary Statistics—Panel Data

Variable	Level	Mean%	Std. Dev.%	Min%	Max%	Obs
Panel A: 3-Month						
3-Month bond return	overall	0.09	0.64	-3.11	5.31	N=744 n=4 T=186
	between	0.10	0.10	0.0002	0.23	
	within		0.63	-3.25	5.17	
Stock return	overall	0.27	4.91	-27.22	16.60	
	between	0.31	0.08	-0.08	0.66	
	within		4.90	-27.21	16.95	
Inflation	overall	1.59	1.74	-2.5	9.1	
	between	1.84	0.84	0.38	2.27	
	within		1.58	-2.78	8.96	
Panel B: 5-Year						
5-Year bond return	overall	0.18	1.18	-9.38	5.31	N=792 n=4 T=198
	between	0.04	0.04	0.13	0.23	
	within		1.18	-9.43	5.30	
Stock return	overall	0.25	4.52	-19.52	16.60	
	between	0.29	0.06	-0.0005	0.66	
	within		4.51	-19.49	16.85	
Inflation	overall	2.43	2.13	-2.1	11.1	
	between	2.81	0.38	2.11	2.95	
	within		2.11	-2.15	10.92	
Panel C: 10-Year						
10-Year bond return	overall	0.26	1.33	-7.44	6.30	N=880 n=5 T=176
	between	0.12	0.12	0.11	0.42	
	within		1.33	-7.46	6.14	
Stock return	overall	0.14	5.53	-28.24	18.76	
	between	0.18	0.05	-0.28	0.67	
	within		5.52	-27.82	19.18	
Inflation	overall	1.86	1.91	-2.5	10.1	
	between	2.75	0.88	0.39	2.53	
	within		1.74	-2.47	9.43	

Table 5.1.16 presents the descriptive statistics for panel data. For the 3-month bond return (Panel A), the average monthly return is modest at 0.09%, with very low volatility

(0.64% overall), consistent with its short-term, low-risk nature. The 5-year bond return (Panel B) shows a higher mean return of 0.18% and greater volatility (1.18%), reflecting the increased interest rate risk associated with longer maturities. The 10-year bond return (Panel C) has the highest average among the three maturities (0.26%) and the largest volatility (1.33%), underscoring the typical risk–return trade-off along the yield curve. Across all panels, stock returns display significantly higher volatility than bonds, with standard deviations exceeding 4.9%, while their means remain modest (ranging from 0.14% to 0.27%), indicative of the higher risk inherent in equities. Inflation remains relatively stable, with averages between 1.59% and 1.86%, and standard deviations below 2.1%, reflecting moderate price fluctuations over the sample periods. Notably, the variability of all variables is predominantly driven by within-entity (time-series) fluctuations rather than between-entity differences, suggesting that temporal dynamics play a more substantial role in shaping returns and inflation than cross-sectional differences.

Table 5.1.17: Correlation Matrix—Panel Data

Panel A: 3-Month			
	3-Month bond return	Stock return	Inflation
3-Month bond return	1.0000		
Stock return	-0.1113	1.0000	
Inflation	-0.0068	-0.1327	1.0000
Panel B: 5-Year			
	5-Year bond return	Stock return	Inflation
5-Year bond return	1.0000		
Stock return	-0.0488	1.0000	
Inflation	-0.1396	-0.0967	1.0000
Panel C: 10-Year			
	10-Year bond return	Stock return	Inflation
10-Year bond return	1.0000		
Stock return	-0.1247	1.0000	
Inflation	-0.0613	-0.1718	1.0000

Table 5.1.17 presents the correlation matrix for panel data. In Panel A, the 3-month bond return shows weak and negative correlations with both stock returns (-0.1113) and inflation

(-0.0068), reflecting the stability of short-term instruments and their limited interaction with riskier or inflation-sensitive assets. In Panel B, the 5-year bond return exhibits a slightly weaker negative correlation with stock returns (-0.0488) compared to the 3-month bond, and a moderately stronger negative correlation with inflation (-0.1396), consistent with medium-term bonds being more exposed to inflation risk. In Panel C, the 10-year bond return displays the strongest negative correlation with stock returns (-0.1247) and a modest negative correlation with inflation (-0.0613), suggesting that longer-term bonds are more inversely related to equities, possibly due to their higher duration and sensitivity to economic conditions. Across all panels, the correlations between stock returns and inflation remain weakly negative (ranging from -0.0967 to -0.1718), indicating that stock returns tend to decline slightly in inflationary periods. Interestingly, the 5-year bond return exhibits the smallest negative correlation with stock returns compared to the 3-month and 10-year bonds. This pattern may reflect the intermediate position of the 5-year maturity, which balances between the liquidity-driven safe-haven behavior of short-term bonds and the duration and inflation sensitivity of long-term bonds. During the sample period, the 5-year bond may have been more influenced by monetary policy expectations and less by risk aversion, resulting in a weaker inverse relationship with equity returns.

5.2 Empirical Specification and Results

We now examine the relationship between bond and stock market returns under different inflation regimes, using inflation as the regime-determining variable. The aim is to assess how bond market behavior responds to stock market movements in varying inflationary environments, and thereby evaluate whether bonds continue to serve as an effective hedge or defensive asset under different macroeconomic conditions.

To enrich our analysis, we implement both a country-by-country time series approach and a panel regression framework. In our baseline specification, we regress bond returns at each available maturity or maturity segment on contemporaneous stock market returns, while allowing for regime switching based on the level of inflation. The regime threshold is determined endogenously, following the threshold regression methodology developed by Hansen (2000) for time series and Hansen (1999) for panel data.

Our benchmark specification is given by the following threshold regression model:

$$RB_{it} = \alpha_i + \beta_1 \cdot RS_{it} \cdot \mathbb{I}(\pi_{it} < \bar{\pi}) + \beta_2 \cdot RS_{it} \cdot \mathbb{I}(\pi_{it} \geq \bar{\pi}) + \gamma' \mathbf{CONTR}_{it} + \varepsilon_{it}, \quad (6)$$

where, in this specification, α_i denotes the country fixed effect in the panel regression (or the country-specific intercept in the time series model), while RB_{it} and RS_{it} represent the

bond and stock returns, respectively, for country i at time t . The variable π_{it} is the inflation rate, and $\bar{\pi}$ denotes the estimated regime threshold that separates low- and high-inflation regimes. The function $\mathbb{I}(\cdot)$ is an indicator function that takes the value 1 when its condition is satisfied and zero otherwise, thereby defining the relevant inflation regime. The term $\mathbf{CONTR}_{it}$ is a vector of control variables, and ε_{it} is the error term.

To test the robustness of our findings, we augment the benchmark model with a set of control variables that may be correlated with stock market returns and could otherwise bias the estimated coefficients if omitted. Specifically, the control vector $\mathbf{CONTR}_{it}$ includes inflation π_{it} ; changes in the VIX index, which capture global stock market risk sentiment (primarily driven by the U.S.); changes in the MOVE index, which reflect volatility in the U.S. Treasury bond market; and changes in industrial production (IP) in country i , which serve as a proxy for domestic economic activity. Among these, only inflation is made to be regime dependent in the robustness checks.

We now display the country by country time series results.

5.3 Time Series Threshold Results

5.3.1 US Results

Table 5.3.1: Threshold Regression Results for R30D Treasury Bill (Sample: 1926M01–2021M10) — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.50%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0002	0.0008	-0.2061	0.8367
C	0.14***	0.01	10.0335	0.0000
<i>Inflation ≥ 2.50%</i>				
<i>(High Regime)</i>				
Stock market	0.0016	0.0030	0.5492	0.5830
C	0.38***	0.03	13.6242	0.0000
<i>Adj R²</i>	2.22%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.2: Threshold Regression Results for Returns on 5Y Bond (Sample: 1926M01–2021M10) — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0034	0.0081	-0.4228	0.6725
C	0.29***	0.05	5.9948	0.0000
<i>Inflation ≥ 3%</i>				
<i>(High Regime)</i>				
Stock market	0.0543***	0.0122	4.4646	0.0000
C	0.08***	0.06	10.5131	0.0000
<i>Adj R²</i>	2.96%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.3: Threshold Regression Results for Returns on R20Y Bond (Sample: 1926M01–2021M10) — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 5.10%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0022	0.0145	-0.1522	0.8791
C	0.50***	0.08	6.2588	0.0000
<i>Inflation ≥ 5.10%</i>				
<i>(High Regime)</i>				
Stock market	0.2140***	0.0347	6.1673	0.0000
C	0.33**	0.16	2.0529	0.0403
<i>Adj R²</i>	2.99%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.1-5.3.3 report the threshold regression results on 30-day, 5-year and 20-year government bond returns separately. For the 30-day Treasury Bill, with an estimated inflation threshold of 2.5%, the stock-bond coefficients remain insignificant in both regimes, though the signs and magnitudes vary. This result suggests that short-term Treasuries exhibit limited sensitivity to stock market fluctuations irrespective of the inflationary environment. Given the high-frequency volatility of short-term securities and their sensitivity to market

liquidity, the residuals in these regressions often exhibit heteroskedasticity and autocorrelation. To address these issues and ensure reliable inference, we employ Newey-West HAC standard errors with a Bartlett kernel and fixed bandwidth. The results underscore the muted comovement between stocks and short-term Treasuries. For the 5-year bond, with a threshold of 3%, the stock-bond correlation remains negative but insignificant in the low-inflation regime, turning significantly positive in the high-inflation regime, consistent with the hypothesis that inflation amplifies stock-bond comovement. For the 20-year bond, with a threshold of 5.1%, the stock-bond correlation is insignificant below the threshold but becomes significantly positive above it, suggesting that long-term bonds lose their hedging properties and move in tandem with stocks when inflation is high.

Table 5.3.4: Threshold Regression Results for Returns on 30-day Treasury Bill (Sample: 1926M01–2021M10) with controls — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.40%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0002	0.0007	-0.2938	0.7689
Inflation	0.01**	0.00	2.3973	0.0167
C	0.14***	0.01	9.9968	0.0000
<i>Inflation ≥ 2.40%</i>				
<i>(High Regime)</i>				
Stock market	0.0022	0.0028	0.7707	0.4410
Inflation	0.02	0.01	1.3131	0.1894
C	0.29***	0.0006	4.9356	0.0000
<i>Adj R²</i>	24.82%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.5: Threshold Regression Results for Returns on 5-year Bond (Sample: 1926M01–2021M10) with controls — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0034	0.0081	-0.4223	0.6729
Inflation	-0.00	0.02	-0.0211	0.9832
C	0.29***	0.05	5.8492	0.0000
<i>Inflation ≥ 3%</i>				
<i>(High Regime)</i>				
Stock market	0.0535***	0.0122	4.4000	0.0000
Inflation	-0.04**	0.02	-2.2179	0.0268
C	0.80***	0.11	6.9837	0.0000
<i>Adj R²</i>	3.21%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.6: Threshold Regression Results for Returns on 20-year Bond (Sample: 1926M01–2021M10) with controls — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3</i>				
<i>(Low Regime)</i>				
Stock market	-0.0140	0.0161	-0.8679	0.3856
Inflation	-0.01	0.04	-0.1468	0.8833
C	0.38***	0.10	3.8396	0.0001
<i>Inflation ≥ 3</i>				
<i>(High Regime)</i>				
Stock market	0.1331***	0.0241	5.5165	0.0000
Inflation	-0.08**	0.03	-2.3784	0.0176
C	1.07***	0.23	4.7092	0.0000
<i>Adj R²</i>	3.10%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.4-5.3.6 report the threshold regression result on 30-day, 5-year and 20-year U.S. bond returns controlling for inflation. The results are consistent with previous ones. For

the 30-day Treasury Bill, there is no significant stock-bond correlation in either regime, indicating that short-term treasuries are less responsive to the stock market movements and inflation changes. For the 5-year and 20-year bond, the coefficients remain negative but insignificant in the low-inflation regimes, while continuously being positive and significant in the high-inflation regime with significantly negative coefficients of inflation for both series, reflecting that part of the comovement was initially capturing direct inflation effects. Collectively, these findings highlight the regime-dependent nature of stock-bond correlations, which intensify and turn positive at higher inflation levels particularly for 5-year and 20-year bonds, while remaining muted for 30-day Treasury securities. In the low-inflation regime, the stock coefficients tend to be negative, suggesting a modest hedging role in which bonds move counter to stocks; however, this relationship is often not statistically significant, yet it does not appear to undermine the diversification benefits of government securities. Bond maturity also plays an important role: intermediate- and long-term bonds exhibit clear regime-dependent correlations with a detected threshold effect, whereas short-term bonds display little sensitivity to shifts in inflation regimes.

5.3.2 Japan Results

Table 5.3.7: Threshold Regression Results for Returns on UISRMJ3E (Sample: 1990M02–2022M08) — Japan

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0017	0.001	-1.6424	0.1013
C	0.013**	0.006	2.1773	0.0000
<i>Inflation ≥ 1.80%</i>				
<i>(High Regime)</i>				
Stock market	0.0141***	0.0018	7.6807	0.0000
C	0.0179**	0.0136	1.3174	0.1885
<i>Adj R²</i>	13.2%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.7 reports the threshold regression results on 3-month bond returns in Japan. The estimated inflation threshold is 1.80%, distinguishing the series between low- and high-inflation regimes. When inflation is below 1.80%, the stock market coefficient is negative

(-0.0017) and statistically insignificant, suggesting that stock and 3-month bond returns are not meaningfully correlated in low-inflation periods. In contrast, when inflation is at or above 1.80%, the stock market coefficient becomes positive (0.0141) and highly significant, indicating a strong positive comovement between stocks and bonds under higher inflation. These results highlight a clear threshold effect: the stock–bond relationship is not stable across regimes but intensifies as inflation rises. The findings imply that inflation plays an important role in shaping the interaction between stock and bond markets, with bonds behaving more like stocks when inflation is high, and more independent when inflation is low.

Table 5.3.8: Threshold Regression Results for Returns on UISRMJ1E (Sample: 1990M02–2022M08) — Japan

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.90%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0551***	0.01	-5.5255	0.0000
<i>C</i>	0.2215***	0.0572	3.8683	0.0001
<i>Inflation ≥ 1.90%</i>				
<i>(High Regime)</i>				
Stock market	0.095***	0.018	5.2755	0.0000
<i>C</i>	0.481***	0.1340	3.5884	0.0004
<i>Adj R²</i>	12.6%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.8 reports the threshold regression results on 10-year bond returns in Japan. The estimated inflation threshold is 1.90%, dividing the sample into low- and high-inflation regimes. When inflation is below 1.90%, the stock market coefficient is negative and highly significant (-0.0551), suggesting that stock and bond returns tend to move in opposite directions under low inflation. This negative comovement aligns with the traditional view of stocks and bonds as substitutes, providing diversification benefits to investors during periods of price stability. However, when inflation rises to 1.90% or higher, the stock market coefficient becomes positive (0.0950) and remains strongly significant, indicating that stocks and bonds start moving together in high-inflation environments. This shift implies that the usual diversification between the two assets breaks down when inflation is elevated, and both markets respond similarly to economic shocks. The results point to a clear threshold effect driven by inflation, during high-inflation regime, the bonds' protection role for stocks vanishes.

Table 5.3.9: Threshold Regression Results for Returns on UISRMJ3E (Sample: 1990M02–2022M08) with controls — Japan

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0017	0.0010	-1.596	0.1113
Inflation	0.0089	0.0082	1.0885	0.2771
C	0.0128**	0.006	2.1461	0.0325
<i>Inflation ≥ 1.80%</i>				
<i>(High Regime)</i>				
Stock market	0.0141***	0.0018	7.7121	0.0000
Inflation	-0.0387*	0.0205	-1.8878	0.0598
C	0.1211**	0.0563	2.1508	0.0321
<i>Adj R²</i>	13.9%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.9 reports the threshold regression result on 3-month Japanese bond returns controlling for inflation. The coefficient of stock market returns remains significant positive in high-inflation regime and insignificant in low-inflation regime with same magnitudes as previous results.

Table 5.3.10: Threshold Regression Results for Returns on UISRMJ1E (Sample: 1990M02–2022M08) with controls — Japan

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.90%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0556***	0.01	-5.5583	0.0000
Inflation	-0.072	0.0768	-0.9381	0.3488
C	0.2245***	0.0574	3.909	0.0001
<i>Inflation ≥ 1.90%</i>				
<i>(High Regime)</i>				
Stock market	0.095***	0.018	5.2683	0.0000
Inflation	-0.038*	0.207	-0.1836	0.8544
C	0.5839	0.5765	1.0129	0.3117
<i>Adj R²</i>	12.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.10 reports the threshold regression result on 10-year Japanese bond returns controlling for inflation. The coefficient of stock market is significant negative in the low-inflation regime and significant positive in the high-inflation regime, which is consistent with earlier findings.

5.3.3 Canada Results

In the following analysis of Canadian bond returns, we group the results into three parts for clarity. First, we examine the Canada 3M Treasury (SPFICT3T) and the Canada Total Government Treasury Bill (SPFICTBT) together, as they both represent short-term, highly liquid instruments. Second, we jointly discuss the Canada Government Bond (SPFCGBT) and the Canada Aggregate Bond (SPFCAGG), given their similar comprehensive nature. Finally, we analyze the Canada Government 1-5Y Bond (SPFCG15T) separately, given its unique maturity profile and distinct dynamics.

Table 5.3.11: Threshold Regression Results for Returns on SPFICT3T (Sample: 2007M03–2023M09) — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0021**	0.0010	-2.1544	0.0324
C	0.0173***	0.0058	3.0138	0.0029
Non-Threshold Variables				
SPFICT3T(-1)	0.8580***	0.3514	24.4135	0.0000
<i>Adj R²</i>	75.9%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.12: Threshold Regression Results for Returns on SPFICTBT (Sample: 2007M03–2023M09) — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0045***	0.0011	-4.0964	0.0001
C	0.0241***	0.0065	3.7351	0.0002
Non-Threshold Variables				
SPFICTBT(-1)	0.8067***	0.0390	20.6692	0.0000
<i>Adj R²</i>	70.35%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.11-5.3.12 report the threshold regression results on government 3-month Treasury and total government T-bill returns in Canada. In the initial static specifications, the residuals exhibited substantial positive serial correlation, as indicated by a very low Durbin-Watson statistic. This suggests that the model failed to capture the dynamic nature of Canadian government bond returns. To address this issue, we included the lagged dependent variable as a non-threshold control variable in the regression. This approach explicitly models the autoregressive structure of returns, substantially reduces serial correlation in the residuals, and improves the explanatory power of the model (Adjusted increasing to approximately 76% for the 3M and 70% for the Total Government Treasury Bill). The estimated coefficients on the lagged dependent variables are large and highly significant (0.86 and 0.81, respectively), confirming the strong inertia present in Canadian government bond returns.

Regarding the regression results of threshold model, there are no thresholds selected in both series, which is likely due to the positive auto-correlation in dependent variables. The coefficients of stock market are both negative and significant. The negative stock-bond correlation indicates that Canadian short-term treasuries served as hedging tools to offset stock market risks.

Table 5.3.13: Threshold Regression Results for Returns on SPFCGBT (Sample: 1998M02–2023M09) — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0220	0.0158	1.3913	0.1651
C	0.3116***	0.0690	4.5141	0.0000
<i>Adj R</i> ²	0.3%			
Significance is denoted by * <i>p</i> < .10; ** <i>p</i> < .05; *** <i>p</i> < .01.				

Table 5.3.14: Threshold Regression Results for Returns on SPFCAGG (Sample: 1998M02–2023M09) — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0305**	0.0148	2.0602	0.0402
C	-0.0405	0.0645	-0.6270	0.5311
<i>Adj R</i> ²	1.04%			
Significance is denoted by * <i>p</i> < .10; ** <i>p</i> < .05; *** <i>p</i> < .01.				

Table 5.3.13-5.3.14 report the threshold regression results on government bond and aggregate bond(both including all maturities; aggregate bond includes government bond, municipal bond, corporate bond and so on) returns in Canada. For both series, no significant threshold is detected, suggesting no evidence of regime-dependent dynamics. For government bond, the stock-bond correlation is positive but insignificant, indicating no correlation between stock and bond returns. For aggregate bond, the stock and bond returns are significantly positively correlated. These findings indicate that Canadian government and aggregate bonds exhibit relatively stable and positive comovement with stocks, with no apparent inflation-driven regime shifts, in contrast to the results observed for shorter-term instruments.

Table 5.3.15: Threshold Regression Results for Returns on SPFCG15T (Sample: 1998M02–2023M09) — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0125	0.0077	-1.6285	0.1045
C	0.2653***	0.0334	7.9452	0.0000
<i>Adj R</i>²	0.5%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.15 reports the threshold regression results on 1-5 years government bond in Canada. No threshold was detected and the coefficient of stock market is negative and insignificant, indicating that the stock-bond correlation does not exist and 1-5 year bonds returns are stably positive regardless of stock market movements.

Table 5.3.16: Threshold Regression Results for Returns on SPFICT3T (Sample: 2007M03–2023M09) with controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0022**	0.0010	-2.2263	0.0271
Inflation	-0.0023	0.0027	-0.8469	0.3981
C	0.0208***	0.0070	2.9521	0.0035
Non-Threshold Variables				
SPFICT3T(-1)	0.8703***	0.0381	22.8615	0.0000
<i>Adj R</i>²	75.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.17: Threshold Regression Results for Returns on SPFICTBT (Sample: 2007M03–2023M09) with controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0046***	0.0011	-4.1894	0.0000
Inflation	-0.0031	0.0029	-1.0761	0.2832
C	0.0294***	0.0081	3.6275	0.0004
Non-Threshold Variables				
SPFICTBT(-1)	0.8198***	0.0409	20.0675	0.0000
<i>Adj R²</i>	70.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.16-5.3.17 report the threshold regression results on government 3-month Treasury and total government T-bill returns in Canada controlling for inflation. Again, no thresholds are detected. Both the 3-month Treasury and total government T-bill returns exhibit strong persistence, with the lagged dependent variable highly significant and accounting for the majority of the variation in returns. The stock market return remains significantly negatively related to bond returns, consistent with their role as safe-haven assets, with a stronger effect for the total government bonds. Inflation shows a negative but statistically insignificant effect, suggesting that short-term fluctuations in inflation had limited direct impact on bond returns during the sample period.

Table 5.3.18: Threshold Regression Results for Returns on SPFCGBT (Sample: 1998M02–2023M09) with controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0189	0.0159	1.1934	0.2336
Inflation	-0.0927*	0.0496	-1.8679	0.0627
C	0.5121***	0.1275	4.0171	0.0001
<i>Adj R²</i>	1.1%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.19: Threshold Regression Results for Returns on SPFCAGG (Sample: 1998M02–2023M09) with controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0275*	0.0148	1.8568	0.0643
Inflation	-0.0889*	0.0464	-1.9167	0.0562
C	0.1519	0.1191	1.2746	0.2034
<i>Adj R</i>²	1.9%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.18-5.3.19 report the threshold regression results on government bond and aggregate bond returns in Canada controlling for inflation. For both series, no significant threshold is detected, suggesting no regime-dependent dynamics. For government bond, the stock-bond correlation remains positive and insignificant, while inflation enters with a marginally significant negative coefficient. For aggregate bond, the stock-bond correlation is marginally positive, while inflation is also marginally negative. These results indicate that Canadian government and aggregate bonds exhibit weakly positive comovement with stocks and modest sensitivity to inflation, but no evidence of regime shifts is observed.

Table 5.3.20: Threshold Regression Results for Returns on SPFCG15T (Sample: 1998M02–2023M09) with controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.5%</i> (<i>Low Regime</i>)				
Stock market	-0.0236**	0.0098	-2.4035	0.0168
Inflation	-0.0014	0.0572	-0.0245	0.9805
C	0.2704***	0.0955	2.8309	0.0050
<i>Inflation ≥ 2.5%</i> (<i>High Regime</i>)				
Stock market	0.0041	0.0117	0.3464	0.7293
Inflation	-0.1814***	0.0430	-4.2190	0.0000
C	0.9698***	0.1741	5.5706	0.0000
<i>Adj R</i>²	5.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.20 reports the threshold regression results on 1-5 years government bond returns in Canada controlling for inflation. A threshold value of 2.5% was detected, dividing the series into two different regimes. In the low-inflation regime, the stock-bond correlation is significantly negative, suggesting that 1-5 year bonds serve as a hedge against stocks when inflation is low. In the high-inflation regime, the stock-bond correlation becomes insignificant, while inflation itself becomes significantly negative, indicating that bond returns deteriorate as inflation rises beyond the threshold. These findings highlight the regime-dependent dynamics of medium-term bonds and the adverse impact of high inflation on their performance.

5.3.4 Eurozone Results

Table 5.3.21: Threshold Regression Results for Returns on EMTT3UC (Sample: 2003M10–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.80%</i>				
<i>(Low Regime)</i>				
Stock market	0.0005	0.0005	1.0047	0.3160
C	0.0033	0.0024	1.4098	0.1598
<i>Inflation ≥ 2.80%</i>				
<i>(High Regime)</i>				
Stock market	-0.0020**	0.0008	-2.5804	0.0104
C	-0.0074	0.0051	-1.4404	0.1510
Non-Threshold Variables				
EMTT3UC(-1)	0.9755***	0.0149	65.6201	0.0000
<i>Adj R²</i>	94.7%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.21 reports the threshold regression results on 0-3 months government bill returns in Eurozone. Due to a extremely low Durbin-Watson value in the initial regression, we use lagged dependent variable as a non-threshold variable. The threshold regression for European 3-month government bond returns identifies a significant threshold, indicating potential regime-dependent dynamics. In the low-inflation regime, the coefficient of stock market is positive but insignificant, indicating there is no stock-bond correlation. However, in the high inflation regime (above 2.8%), the relationship becomes significantly negative,

although slightly (coefficient=-0.0020), indicating that higher stock market returns are associated with lower short-term bond returns. This negative association likely reflects tighter monetary policy in response to elevated inflation, where rising short-term interest rates depress stock valuations while increasing bond yields. Additionally, heightened risk aversion during high inflation may lead investors to shift from equities to safer short-term government securities, reinforcing the inverse relationship.

Table 5.3.22: Threshold Regression Results for Returns on EXEA5 (Sample: 2007M03–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0012	0.0059	0.2010	0.8409
C	0.1011**	0.0306	3.3016	0.0011
<i>Adj R²</i>	-0.45%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.22 reports the threshold regression results on 1-3 years government bond returns in Eurozone. The initial threshold regression does not detect a significant threshold, suggesting no evidence of regime-dependent dynamics over the sample period. The stock–bond relationship remains statistically insignificant, with a negligible coefficient and a high p-value. The model exhibits almost no explanatory power, as reflected by the near-zero *Adj R²*.

Table 5.3.23: Threshold Regression Results for Returns on EXEB5 (Sample: 2007M03–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.50%</i> (<i>Low Regime</i>)				
Stock market	0.0242*	0.0130	1.8615	0.0641
C	0.1758***	0.0637	2.7602	0.0063
<i>2.50% ≤ Inflation < 3.30%</i> (<i>Mid Regime</i>)				
Stock market	-0.0740***	0.0234	-3.1582	0.0018
C	0.3983***	0.1424	2.7971	0.0056
<i>Inflation ≥ 3.30%</i> (<i>High Regime</i>)				
Stock market	0.0661**	0.0255	2.5952	0.0101
C	-0.2411*	0.1371	-1.7588	0.0801
<i>Adj R²</i>	12.7%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.23 reports the threshold regression results on 3-5 years government bond returns in Eurozone. The results show that two significant thresholds were detected (with values of 2.5% and 3.3%), and separate the stock-bond correlation into three regimes. In the low-inflation regime ($\text{Inflation} < 2.50\%$), the stock-bond correlation is marginally positive at 10% level. In the intermediate regime ($2.50\% \leq \text{Inflation} < 3.30\%$), the relationship turns significantly negative, indicating that a strong hedging behavior. In the high-inflation regime ($\text{Inflation} \geq 3.30\%$), the relationship again becomes significantly positive, indicating that extreme inflation leads to a breakdown of the traditional stock-bond hedge and a strong comovement.

Table 5.3.24: Threshold Regression Results for Returns on EXEC5 (Sample: 2004M09–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.70%</i>				
<i>(Low Regime)</i>				
Stock market	0.0149	0.0173	0.8627	0.3892
C	0.3368***	0.0807	4.1762	0.0000
<i>Inflation ≥ 1.70%</i>				
<i>(High Regime)</i>				
Stock market	0.0197	0.0296	0.6642	0.5072
C	-0.3636**	0.1825	-1.9925	0.0474
Adj R²	4.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.24 reports the threshold regression results on 5-7 years government bond returns in Eurozone. the threshold regression identifies a significant inflation threshold at 1.7%, dividing the sample into low- and high-inflation regimes. The stock-bond relationship remains statistically insignificant in both regimes.

Table 5.3.25: Threshold Regression Results for Returns on EXED5 (Sample: 2008M01–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.40%</i>				
<i>(Low Regime)</i>				
Stock market	0.0151	0.0225	0.6708	0.5031
C	0.4031***	0.1171	3.4426	0.0007
<i>Inflation ≥ 3.40%</i>				
<i>(High Regime)</i>				
Stock market	0.2055***	0.0517	3.9761	0.0001
C	-0.5163*	0.2821	-1.8305	0.0686
Adj R²	11.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.25 reports the threshold regression results on 7-10 years government bond returns in Eurozone. the threshold regression identifies a significant inflation threshold at 3.4%,

dividing the sample into low- and high-inflation regimes. In the low-inflation regime, the stock-bond relationship is insignificant, and bond returns remain positive. However, in the high-inflation regime, the stock-bond relationship becomes significantly positive, suggesting stronger comovement between stocks and bonds when inflation is high.

Table 5.3.26: Threshold Regression Results for Returns on EXEE5 (Sample: 2007M03–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.30%</i>				
<i>(Low Regime)</i>				
Stock market	0.0498*	0.0294	1.6920	0.0921
C	0.4581***	0.1504	3.0461	0.0026
<i>Inflation ≥ 3.30%</i>				
<i>(High Regime)</i>				
Stock market	0.2801***	0.0663	4.2240	0.0000
C	-0.7354**	0.3566	-2.0619	0.0404
<i>Adj R²</i>	12.6%			
Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.				

Table 5.3.26 reports the threshold regression results on 10-15 years government bond returns in Eurozone. The model identifies a significant inflation threshold at 3.30%. In the low-inflation regime, the stock-bond correlation is marginally positive at 10% level. In the high-inflation regime, the stock-bond correlation becomes strongly and significantly positive, suggesting intensified comovement between stocks and bonds.

Table 5.3.27: Threshold Regression Results for Returns on EXIEF5 (Sample: 2004M02–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.10%</i> (<i>Low Regime</i>)				
Stock market	0.0222	0.0465	0.4773	0.6336
C	0.5206**	0.2155	2.4155	0.0164
<i>Inflation ≥ 3.10%</i> (<i>High Regime</i>)				
Stock market	0.2614***	0.0855	3.0591	0.0025
C	-0.9916*	0.5316	-1.8652	0.0633
<i>Adj R²</i>	6.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.27 reports the threshold regression results on 15-year and above government bond returns in Eurozone. The model identifies a significant inflation threshold at 3.10%, dividing the sample into low- and high-inflation regimes. In the low-inflation regime, the stock-bond correlation remains insignificant. In the high-inflation regime, the stock-bond correlation becomes strongly and significantly positive. These results suggest that elevated inflation intensifies stock-bond comovement and undermines the diversification benefits of long-term bonds.

Table 5.3.28: Threshold Regression Results for Returns on EMTT3UC (Sample: 2003M10–2025M03) with controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0001	0.0004	-0.3334	0.7391
Inflation	-0.0025**	0.0010	-2.3914	0.0175
C	0.0068**	0.0030	2.2910	0.0228
Non-Threshold Variables				
EMTT3UC(-1)	0.9813***	0.0149	65.7487	0.0000
<i>Adj R²</i>	94.7%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.28 reports the threshold regression results on 0-3 months government bill returns

in Eurozone controlling for inflation. After including inflation as an explanatory variable, the threshold regression for European 3-month government bill returns fails to identify a significant threshold, suggesting that the dynamics of returns are adequately captured by the included variables without evidence of regime shifts. The correlation between stock and bond returns turns to insignificant, while the coefficient of inflation is significantly negative. This indicates that while stock returns have limited explanatory power for 3-month bond returns, inflation plays a more prominent role in high-inflation environments.

Table 5.3.29: Threshold Regression Results for Returns on EXEA5 (Sample: 2007M03–2025M03) with controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.10%</i>				
<i>(Low Regime)</i>				
Stock market	0.0075	0.0077	0.9752	0.3306
Inflation	-0.0369	0.0529	-0.6980	0.4859
C	0.1083	0.0621	1.7446	0.0825
<i>Inflation ≥ 2.10%</i>				
<i>(High Regime)</i>				
Stock market	-0.0045	0.0086	-0.5172	0.6056
Inflation	-0.0914***	0.0204	-4.4817	0.0000
C	0.4937***	0.0928	5.3212	0.0000
<i>Adj R²</i>	7.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.29 reports the threshold regression results on 1-3 years government bond returns in Eurozone controlling for inflation. The threshold regression for European 1-3 year government bond returns identifies a significant threshold after adding inflation as an explanatory variable. In the high-inflation regime, inflation has a strong and statistically significant negative effect on bond returns, consistent with the notion that higher inflation erodes fixed-income assets. In contrast, stock returns remain insignificant in both regimes, and inflation does not appear to matter in the low-inflation regime. It suggests that inflation plays a more prominent role in driving bond returns when it is beyond a certain threshold at 2.1%, while the stock-bond correlation remains weak.

Table 5.3.30: Threshold Regression Results for Returns on EXEB5 (Sample: 2007M03–2025M03) with controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.50%</i>				
<i>(Low Regime)</i>				
Stock market	0.0239*	0.0129	1.8537	0.0652
Inflation	-0.0360	0.0763	-0.4723	0.6372
C	0.2164**	0.1067	2.0290	0.0437
<i>2.50% ≤ Inflation < 3.30%</i>				
<i>(Mid Regime)</i>				
Stock market	-0.0764***	0.0242	-3.1643	0.0018
Inflation	-0.2410	0.6728	-0.3582	0.7206
C	1.0619	1.8583	0.5715	0.5683
<i>Inflation ≥ 3.30%</i>				
<i>(High Regime)</i>				
Stock market	0.0847***	0.0262	3.2372	0.0014
Inflation	-0.1628***	0.0611	-2.6642	0.0083
C	0.7803*	0.4067	1.9189	0.0564
<i>Adj R²</i>	14.5%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.30 reports the threshold regression results on 3-5 years government bond returns in Eurozone controlling for inflation. The threshold regression continues to identify two significant thresholds, yielding three regimes. The stock-bond relationship remains non-linear, with a marginally positive correlation in the low-inflation regime, a significant negative correlation in the intermediate regime, and a significant positive correlation in the high-inflation regime. Inflation becomes significantly negative only in the high-inflation regime, consistent with earlier findings.

Table 5.3.31: Threshold Regression Results for Returns on EXEC5 (Sample: 2004M09–2025M03) with controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0208	0.0148	1.4024	0.1621
Inflation	-0.1237***	0.0364	-3.3972	0.0008
C	0.4813***	0.1069	4.5009	0.0000
<i>Adj R</i>²	4.7%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.31 reports the threshold regression results on 5-7 years government bond returns in Eurozone controlling for inflation. The model no longer detects a significant threshold after adding inflation as an explanatory variable, suggesting that the regime-dependent dynamics observed earlier is largely explained by inflation itself. Inflation enters the model with a significant and negative effect on bond returns, consistent with the idea that higher inflation reduces the value of fixed-income securities. This result indicates that the threshold effect previously detected reflects the non-linear impact of inflation, which becomes directly captured once inflation is explicitly controlled for.

Table 5.3.32: Threshold Regression Results for Returns on EXED5 (Sample: 2008M01–2025M03) with controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.40%</i> (<i>Low Regime</i>)				
Stock market	0.0145	0.0224	0.6452	0.5195
Inflation	-0.0304	0.1175	-0.2589	0.7960
C	0.4452**	0.1996	2.2300	0.0269
<i>Inflation ≥ 3.40%</i> (<i>High Regime</i>)				
Stock market	0.2530***	0.0540	4.6860	0.0000
Inflation	-0.3511***	0.1305	-2.6898	0.0078
C	1.7639**	0.8923	1.9769	0.0494
<i>Adj R</i>²	13.6%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.32 reports the threshold regression results on 7-10 years government bond returns in Eurozone controlling for inflation. The model continuously detect a significant threshold at 3.40%. In the low-inflation regime, neither stock returns nor inflation significantly affect bond returns, while the intercept remains positive and significant. In the high-inflation regime, stock returns show a strong and significant positive relationship with bond returns, and inflation exhibits a significant negative effect, suggesting that elevated inflation diminishes bond performance and strengthens stock-bond comovement.

Table 5.3.33: Threshold Regression Results for Returns on EXEE5 (Sample: 2007M03–2025M03) with controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.30%</i>				
<i>(Low Regime)</i>				
Stock market	0.0471	0.0294	1.6029	0.1105
Inflation	-0.1180	0.1540	-0.7660	0.4445
C	0.6240**	0.2630	2.3731	0.0185
<i>Inflation ≥ 3.30%</i>				
<i>(High Regime)</i>				
Stock market	0.3214***	0.0682	4.7127	0.0000
Inflation	-0.3622**	0.1593	-2.2734	0.0240
C	1.5364	1.0599	1.4495	0.1487
<i>Adj R²</i>	13.6%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.33 reports the threshold regression results on 10-15 years government bond returns in Eurozone controlling for inflation. The threshold regression for European 10-15year bond returns continues to detect a significant threshold at 3.30%, confirming the regime-dependent dynamics. In the low-inflation regime, stock returns and inflation remain insignificant. In the high-inflation regime, stock returns exhibit a strong and significant positive association with bond returns, and inflation has a significant negative effect, indicating that elevated inflation erodes bond returns and amplifies stock-bond comovement.

Table 5.3.34: Threshold Regression Results for Returns on EXIEF5 (Sample: 2004M09–2025M03) with controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0876**	0.0405	2.1645	0.0314
Inflation	-0.3765***	0.1000	-3.7673	0.0002
C	1.0345***	0.2917	3.5463	0.0005
<i>Adj R²</i>	6.6%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.34 reports the threshold regression results on 15y and above government bond returns in Eurozone controlling for inflation. The threshold regression no longer detects threshold, suggesting that the previously observed regime-dependent dynamics are largely explained by inflation itself. Both stock returns and inflation remain significant, with stock returns showing a positive association with bond returns and inflation exerting a significant negative effect.

5.3.5 UK Results

Table 5.3.35: Threshold Regression Results for Returns on FTRFBGL (Sample: 2004M04–2025M03) — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0016***	0.0100	-0.1613	0.8720
C	0.1955***	0.0372	5.2544	0.0000
<i>Adj R²</i>	-0.03%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.36: Threshold Regression Results for Returns on FTRFBGM (Sample: 2004M04–2025M03) — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0073	0.0307	0.2387	0.8115
C	0.2589**	0.114	2.2705	0.024
<i>Adj R²</i>	-0.03%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.37: Threshold Regression Results for Returns on FTRFBGM1 (Sample: 1999M08–2025M03) — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0167	0.0227	-0.7337	0.4637
C	0.3041***	0.0886	3.4326	0.0007
<i>Adj R²</i>	-0.02%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.38: Threshold Regression Results for Returns on FTRFBGM2 (Sample: 1999M08–2025M03) — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.014	0.0318	-0.4397	0.6605
C	0.3116**	0.1242	2.5083	0.0126
<i>Adj R²</i>	-0.03%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.39: Threshold Regression Results for Returns on FTRFBGH (Sample: 1999M08–2025M03) — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0012	0.0488	0.0236	0.9812
C	0.2501	0.1903	1.3143	0.1897
<i>Adj R</i>²	-0.03%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.35-5.3.39 report the threshold regression results on 0-5 years, 5-15 years, 5-10 years, 10-15 years and 15-year and above government bond returns in the UK. The initial regressions across all maturities, before controlling for inflation and other variables, consistently fail to detect a significant inflation threshold. The stock–bond relationship is statistically insignificant in all cases. The *Adj R*² is effectively zero and slightly negative (-0.03% to -0.02%), suggesting that the model explains none of the variation in bond returns and performs worse than a simple mean prediction. These results indicate that there is no evidence of a meaningful stock–bond correlation or regime-dependent dynamics in the UK within the full sample.

Table 5.3.40: Threshold Regression Results for Returns on FTRFBGL (Sample: 2004M04–2025M03) with controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 4%</i> (<i>Low Regime</i>)				
Stock market	-0.0035	0.0113	-0.3115	0.7557
Inflation	-0.0029	0.0424	-0.0691	0.945
C	0.1975**	0.0944	2.0921	0.0374
<i>Inflation ≥ 4%</i> (<i>High Regime</i>)				
Stock market	0.0175	0.0206	0.8473	0.3976
Inflation	-0.1381***	0.036	-3.8365	0.0002
C	1.1358***	0.2537	4.4775	0.0000
<i>Adj R</i>²	3.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.41: Threshold Regression Results for Returns on FTRFBGM (Sample: 2004M04–2025M03) with controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 4%</i> (<i>Low Regime</i>)				
Stock market	-0.0289	0.0342	-0.8447	0.3991
Inflation	-0.0417	0.1285	-0.3247	0.7457
C	0.3741	0.2863	1.3067	0.1925
<i>Inflation ≥ 4%</i> (<i>High Regime</i>)				
Stock market	0.1576**	0.0625	2.52	0.0124
Inflation	-0.4336***	0.1092	-3.9716	0.0001
C	3.1269***	0.7694	4.0638	0.0001
<i>Adj R²</i>	5.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.42: Threshold Regression Results for Returns on FTRFBGM1 (Sample: 1999M08–2025M03) with controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.80%</i> (<i>Low Regime</i>)				
Stock market	-0.0454*	0.0244	-1.866	0.063
Inflation	-0.0769	0.1026	-0.7491	0.4544
C	0.462**	0.2081	2.2204	0.0271
<i>Inflation ≥ 3.80%</i> (<i>High Regime</i>)				
Stock market	0.1497**	0.0521	2.8732	0.0044
Inflation	-0.3916***	0.0914	-4.2839	0.0000
C	2.8822***	0.6356	4.5347	0.0000
<i>Adj R²</i>	6.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.43: Threshold Regression Results for Returns on FTRFBGM2 (Sample: 1999M08–2025M03) with controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0555	0.0341	-1.6252	0.1052
Inflation	-0.1018	0.1438	-0.7077	0.4797
C	0.534*	0.2916	1.8313	0.068
<i>Inflation ≥ 3.80%</i>				
<i>(High Regime)</i>				
Stock market	0.2205***	0.073	3.0192	0.0028
Inflation	-0.5497***	0.1281	-4.2912	0.0000
C	3.8653***	0.8908	4.3393	0.0000
<i>Adj R²</i>	6.8%			
Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.				

Table 5.3.44: Threshold Regression Results for Returns on FTRFBGH (Sample: 1999M08–2025M03) with controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0527	0.0522	-1.0095	0.3135
Inflation	-0.1933	0.2199	-0.8787	0.3803
C	0.7257	0.446	1.6271	0.1048
<i>Inflation ≥ 3.80%</i>				
<i>(High Regime)</i>				
Stock market	0.3026***	0.1117	2.7089	0.0071
Inflation	-0.8829***	0.1959	-4.5062	0.0000
C	5.5208***	1.3624	4.0523	0.0001
<i>Adj R²</i>	7.1%			
Significance is denoted by * <i>p</i> < .10; ** <i>p</i> < .05; *** <i>p</i> < .01.				

Table 5.3.40-5.3.44 report the threshold regression results on 0-5 years, 5-15 years, 5-10 years, 10-15 years and 15y and above government bond returns in the UK controlling

for inflation. The results reveal significant inflation thresholds across all maturities, indicating that there exist structural changes in regime switching process, which is consistent with the theoretical model's prediction of regime-dependent stock-bond dynamics. Inflation is significant negatively correlated with bond returns in high regime across all maturities. Short-term bonds (0-5 years) show no significant relationship in either regime, reflecting their strong anchoring to monetary policy rates and low sensitivity to inflation expectations. In contrast, mid-term bonds (5-10 years) exhibit a slightly significant negative correlation in the low-inflation regime at 10% level, suggesting traditional diversification benefits, but switch to a significant positive correlation under high inflation. Long-term bonds (5-15 years, 10-15 years and 15 year and above) show no significant relationship in low inflation but become significantly and positively correlated with stocks when inflation exceeds the threshold. These patterns align with the theoretical expectation that high inflation amplifies stock-bond comovement, particularly for longer maturities exposed to inflation and term premium risks.

5.3.6 Switzerland Results

Table 5.3.45: Threshold Regression Results for Returns on SPFISWUT (Sample: 2004M02–2025M03) — Switzerland

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0064	0.0248	-0.257	0.7974
C	0.1571*	0.0883	1.7795	0.0764
<i>Adj R²</i>	-0.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.45 reports the threshold regression results of stock returns on government bond returns in Switzerland. The results are insignificant, with no clear threshold identified and weak or inconsistent coefficients, suggesting that the data do not exhibit a meaningful relationship between inflation regimes and the stock-bond correlation, which explains the negative *Adj R²* regarding this model.

Table 5.3.46: Threshold Regression Results for Returns on SPFISWUT (Sample: 2004M02–2025M03) with control — Switzerland

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.007	0.0251	-0.2776	0.7815
Inflation	-0.0125	0.0811	-0.1545	0.8774
C	0.1644	0.1002	1.641	0.102
<i>Adj R²</i>	-0.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.46 reports the threshold regression results on government bond returns in Switzerland controlling for inflation. The results remain the same as the previous, indicating that no threshold is detected and a non-existent correlation of stock-bond with a negative *Adj R²*. The insignificant results for Switzerland, including the failure to identify a threshold, likely reflect the unique economic and financial characteristics of the country. Inflation in Switzerland has been persistently low and stable during the sample period, leaving little room for clear shifts between the low- and high-inflation regimes. Moreover, the Swiss stock market is dominated by multinational firms whose earnings are more sensitive to global economic conditions than to domestic inflation, while the bond market is relatively small and illiquid. Swiss franc assets are also widely regarded as safe havens, and their prices tend to be driven by international capital flows and risk sentiment rather than domestic inflation dynamics. Together, these factors weaken the link between inflation and the stock–bond relationship in Switzerland, making it difficult for the model to detect significant thresholds or correlations.

5.3.7 China Results

Table 5.3.47: Threshold Regression Results for Returns on SPBCN1BT (Sample: 2007M01–2025M03) — China

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0466***	0.0114	-4.0876	0.0001
C	0.3616***	0.0827	4.3736	0.0000
<i>Adj R²</i>	6.7%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.47 reports the threshold regression results on 10-year government bond returns in China. The threshold regression does not identify a significant threshold, suggesting that the stock-bond relationship does not differ substantially across inflation regimes in the sample period. The estimated stock market coefficient is significantly negative, which is consistent with the traditional view of bonds serving as a hedge against equity risk. The *Adj R²* is about 6.7%, suggesting that the model explains a modest proportion of the variation in bond returns.

Table 5.3.48: Threshold Regression Results for Returns on SPBCNGOT (Sample: 2007M01–2025M03) — China

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0026	0.0055	-0.4798	0.6318
C	0.1931***	0.0439	4.4042	0.0000
Non-Threshold Variables				
SPBCNGOT(-1)	0.4343***	0.0618	7.0284	0.0000
<i>Adj R²</i>	19.2%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.48 reports the threshold regression results on government bond(all maturities) returns in China. We conduct lagged dependent variable to address low Durbin-Watson. The regression does not identify a threshold, indicating no clear inflation-driven regime change. The stock market coefficient is negative(-0.0109) but insignificant, suggesting non-exist relationship between stock and bond returns.

Table 5.3.49: Threshold Regression Results for Returns on SPBCN1BT (Sample: 2007M01–2025M03) with control — China

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.4685***	0.0116	-4.026	0.0001
Inflation	-0.0055	0.0427	-0.1276	0.8986
C	0.3741***	0.1285	2.9101	0.004
<i>Adj R</i>²	6.3%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.49 reports the threshold regression results on 10-year government bond returns in China controlling for inflation. The regression results for China's 10-year government bond returns remain largely consistent. No significant threshold is identified and the stock market coefficient remains significantly negative(-0.4685), suggesting that bond returns still move inversely with stock returns, consistent with the traditional hedging role of bonds. The Adjusted is about 6.3%, slightly lower than the model without the control, indicating that the explanatory power of the model remains modest.

Table 5.3.50: Threshold Regression Results for Returns on SPBCNGOT (Sample: 2007M01–2025M03) with control — China

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.5%</i>				
Stock market	0.0135	0.0108	1.2486	0.2132
Inflation	0.2517***	0.0767	3.2834	0.0012
C	0.2145***	0.0735	2.9197	0.0039
<i>1.5% ≤ Inflation < 2.5%</i>				
Stock market	-0.0073	0.0094	-0.7719	0.4410
Inflation	0.6110***	0.2136	2.8609	0.0047
C	-1.0076**	0.4212	-2.3924	0.0176
<i>2.5% ≤ Inflation < 4.30%</i>				
Stock market	-0.0257**	0.0105	-2.4566	0.0149
Inflation	0.5932***	0.1706	3.4770	0.0006
C	-1.6508***	0.5141	-3.2107	0.0015
<i>Inflation ≥ 4.30%</i>				
Stock market	0.0182*	0.0098	1.8636	0.0638
Inflation	0.1034	0.0758	1.3639	0.1741
C	-0.5228	0.4523	-1.1560	0.2490
Non-Threshold Variables				
SPBCNGOT(-1)	0.4783***	0.0592	8.0785	0.0000
<i>Adj R²</i>	33.0%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.3.50 reports the threshold regression results on government bond(all maturities) returns in China controlling for inflation. Surprisingly, there are three threshold values detected, dividing the series into four regimes. In the first two regimes(Inflation <1.5% and 1.5%≤Inflation <2.5%), the stock-bond correlation is weak, with the inflation itself highly positively related to the bond returns. In the regime where 2.5%≤Inflation<4.30%, the stock returns is negatively correlated with the bond returns and the bonds served as a ‘safe heaven’. In the highest regime(Inflation≥4.30%), the coefficient of stock market becomes weakly significantly positive at 10% level, indicating that bonds’ hedging properties gradually disappear, and the inflation has no correlation with the bond market.

5.4 Panel Threshold Results

We employ the Panel Threshold Regression (PTR) model developed by Hansen (1999) to examine whether the relationship between stock returns and bond returns differs across regimes defined by the level of inflation. This method allows us to endogenously determine the threshold value of inflation that splits the sample into distinct regimes, and to estimate the differential effects of explanatory variables across these regimes while controlling for unobserved individual heterogeneity through fixed effects. Bootstrap procedures are used to test the statistical significance of the threshold effect and to construct confidence intervals for the estimated threshold.

Table 5.4.1: Threshold Regression Results for Panel Data

	Panel A	Panel B	Panel C
	3-month	5-year	10-year
Threshold Value $\gamma\%$	7.0000 (0.3500)	5.7000** (0.0200)	7.0000** (0.0133)
<i>Inflation < $\gamma\%$</i> (<i>Low Regime</i>)			
Stock Market	-0.0196*** (0.0000)	-0.0387*** (0.0000)	-0.0415*** (0.0000)
<i>Inflation $\geq \gamma\%$</i> (<i>High Regime</i>)			
Stock Market	0.1119*** (0.0000)	0.2070*** (0.0000)	0.1202*** (0.0000)
C	0.1041*** (0.0000)	0.1929*** (0.0000)	0.2819*** (0.0000)
<i>Within R^2</i>	4.66%	8.45%	4.49%

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.4.2: Threshold Regression Results for Panel Data with control

	Panel A	Panel B	Panel C
	3-month	5-year	10-year
Threshold Value $\gamma\%$	7.4000 (0.3833)	6.9000*** (0.0033)	5.4000*** (0.0067)
<i>Inflation < $\gamma\%$</i> (Low Regime)			
Stock Market	-0.192*** (0.0000)	-0.0354*** (0.0000)	-0.0408*** (0.0000)
Inflation	0.0185 (0.2890)	0.0060 (0.8300)	0.0305 (0.3820)
<i>Inflation $\geq \gamma\%$</i> (High Regime)			
Stock Market	0.0765*** (0.0005)	0.2350*** (0.0000)	0.0557* (0.0570)
Inflation	-0.0796*** (0.0000)	-0.1265*** (0.0000)	-0.1372*** (0.0000)
C	0.0890*** (0.0100)	0.2442*** (0.0001)	0.2769*** (0.0000)
Within R^2	6.95%	13.03%	7.02%

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.4.1-5.4.2 show the threshold regression results on panel data before and after controlling for inflation separately. For the 3-month panel, the estimated threshold for inflation lies approximately around 7%, diving the sample into low- and high-inflation regimes. In the low regime, the stock-bond correlation is negative and statistically significant, reflecting the traditional role of short-term bonds as a hedge against equity risk. In contrast, in the high-inflation regime, the correlation turns positive and remains significant, indicating that short-term bonds lose their hedging property and begin to move in the same direction as stocks in response to inflationary shocks. However, the formal test of the threshold effect yields a p-value above conventional significance levels, suggesting that the difference between the two regimes is not statistically significant. This implies that while the coefficients in the two regimes exhibit opposite signs consistent with theoretical predictions the evidence for a distinct inflation-driven regime shift at the short end of the yield curve is weak. For both

5-year and 10-year panel, the inflation threshold significantly divide the samples into two different regimes with estimated threshold values equal to 5.7% and 7% separately. In the low-inflation regime, the stock-bond correlations are significantly negative, confirming the traditional role of medium- and long-term bonds as effective hedges against equity risk when inflation is contained. In the high-inflation regime, the correlation becomes significantly positive, reflecting the erosion of real returns on bonds and the tendency of both asset classes to move in the same direction in response to inflationary shocks. After controlling for inflation as dependent variable, the main findings remain the same.

5.5 Robustness Checks

In the following section, we will display the robustness checks for time-series results of the seven countries and for panel data results.

5.5.1 US

For the U.S., we conduct an additional analysis using more recent bond return data from Bloomberg, which covers U.S. Treasury securities without distinguishing by maturity. This additional check is particularly relevant because the baseline U.S. regressions rely on data that only extend through 2021, which may not fully reflect the most recent market dynamics. This robustness check serves two purposes. First, it validates whether the regime-dependent stock-bond dynamics identified in our baseline results persist when using an alternative and more contemporary data source. Second, it examines whether the threshold effect holds at the aggregate level, abstracting from differences across maturities. By leveraging Bloomberg’s comprehensive and up-to-date data, we aim to ensure that our conclusions are not specific to a particular sample period or maturity segmentation, thereby enhancing the generalizability of our results.

Table 5.5.1: Threshold Regression Results on Bloomberg Data for Robustness (Sample: 1973M02–2025M02) — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3%</i> (<i>Low Regime</i>)				
Stock market	-0.0455**	0.0209	-2.1736	0.0301
C	0.41***	0.09	4.5783	0.0000
<i>Inflation ≥ 3%</i> (<i>High Regime</i>)				
Stock market	0.0874***	0.0173	5.0647	0.0000
C	0.61***	0.81	7.5818	0.0000
<i>Adj R²</i>	5.10%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.2: Threshold Regression Results on Bloomberg Data for Robustness (Sample: 1973M02–2025M02) with control — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3%</i> (<i>Low Regime</i>)				
Stock market	-0.0451**	0.0210	-2.1501	0.0319
Inflation	0.03	0.10	0.2578	0.7966
C	0.36*	0.21	1.7692	0.0773
<i>Inflation ≥ 3%</i> (<i>High Regime</i>)				
Stock market	0.0864***	0.0173	5.0002	0.0000
Inflation	-0.03	0.03	-1.1868	0.2358
C	0.80***	0.18	4.4702	0.0000
<i>Adj R²</i>	4.98%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.3: Threshold Regression Results on Bloomberg Data for Robustness (Sample: 1973M02–2025M02) with additional controls — U.S.

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.10%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0606**	0.0242	-2.5038	0.0127
Inflation	0.06	0.08	0.7624	0.4462
C	0.30*	0.17	1.7097	0.0881
<i>Inflation ≥ 3.10%</i>				
<i>(High Regime)</i>				
Stock market	0.0748***	0.0281	2.6597	0.0081
Inflation	-0.23***	0.07	-3.1257	0.0019
C	0.0152***	0.0035	4.3614	0.0000
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.05**	0.02	2.5630	0.0107
D(MOVE)	-0.02***	0.01	-4.0785	0.0001
D(IP)	-0.04**	0.02	-2.3210	0.0208
<i>Adj R²</i>	11.39%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.1-5.5.3 show the robustness checks results for U.S. government bond using more recent Bloomberg data. The results again strengthen the main findings. The inflation threshold effect becomes more pronounced and remains significant across specifications. In the high-inflation regime, stock-bond correlations are consistently positive and stronger, confirming the regime-dependent dynamics. The control variables behave as expected: VIX is positively associated with bond returns, reflecting ‘flight-to-safety’ behavior; MOVE is negatively associated, as higher volatility raises yields and lowers bond prices; industrial production growth is negatively related, indicating reduced bond demand when the economy strengthens. Notably, the stock-bond correlation remains significantly negative in the low-inflation regime but turns significantly positive in the high-inflation regime, underscoring the importance of inflation regimes in portfolio decisions. Together, these results support the theoretical prediction that stock-bond correlations increase and turn positive as inflation rises.

5.5.2 Japan

Table 5.5.4: Threshold Regression Results on UISRMJ3E for Robustness (Sample: 1990M02–2022M08) with additional controls — Japan

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0013	0.0012	-1.1062	0.2694
Inflation	0.0084	0.0082	1.0266	0.3052
C	0.0128**	0.006	2.135	0.0334
<i>Inflation ≥ 1.80%</i>				
<i>(High Regime)</i>				
Stock market	0.0142***	0.0019	7.7286	0.0000
Inflation	-0.0373*	0.0207	-1.8074	0.0715
C	0.1172**	0.0566	2.0702	0.0391
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0007	0.0014	0.4711	0.6378
D(MOVE)	0.0002	0.0004	0.5368	0.5917
D(IP)	0.0001	0.005	0.0126	0.99
<i>Adj R²</i>	13.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.5: Threshold Regression Results on UISRMJIE for Robustness (Sample: 1990M02–2022M08) with additional controls — Japan

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.90%</i>				
<i>(Low Regime)</i>				
Stock market	-0.06***	0.0111	-5.3893	0.0000
Inflation	-0.0771	0.0773	-0.9968	0.3195
C	0.2308***	0.0578	3.9964	0.0001
<i>Inflation ≥ 1.90%</i>				
<i>(High Regime)</i>				
Stock market	0.0913***	0.0183	5.0006	0.0000
Inflation	-0.0735	0.2083	-0.3531	0.7242
C	0.6533	0.5788	1.1288	0.2597
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	-0.0132	0.0133	-0.9897	0.3229
D(MOVE)	-0.0005	0.0004	-0.1132	0.91
D(IP)	0.0551	0.048	1.1478	0.2518
<i>Adj R²</i>	12.3%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Tables 5.5.4 and 5.5.5 show the robustness checks results for 3-month and 10-year Japanese bond. The threshold effect of inflation remains stable with the results reveal a significant positive stock-bond correlation in the high-inflation regime for both bond maturities. In low-inflation regime, the stock-bond correlation of 3-month bond is still weak but for 10-year bond, stock and bond returns are positively correlated, showing that long-term bonds serve as better ‘safe haven’ than short-term ones when inflation is low. What’s more, D(VIX), D(MOVE) and D(IP) tend to have no correlation with bond returns in Japan.

5.5.3 Canada

Table 5.5.6: Threshold Regression Results on SPFICT3T for Robustness (Sample: 2007M03–2023M09) with additional controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>No threshold selected</i>				
Stock market	-0.0048***	0.0013	-3.6636	0.0003
Inflation	-0.0031	0.0026	-1.1984	0.2322
C	0.0225***	0.0068	3.2834	0.0012
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
SPFICT3T(-1)	0.8755***	0.0371	23.5720	0.0000
D(VIX)	-0.0035***	0.0010	-3.5846	0.0004
D(MOVE)	0.0006**	0.0003	2.2590	0.0250
D(IP)	-0.0003	0.0009	-0.3122	0.7553
<i>Adj R</i>²	77.22%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.7: Threshold Regression Results on SPFICTBT for Robustness (Sample: 2007M03–2023M09) with additional controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>No threshold selected</i>				
Stock market	-0.0084***	0.0015	-5.5795	0.0000
Inflation	-0.0037	0.0029	-1.3045	0.1936
C	0.0297***	0.0079	3.7505	0.0002
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
SPFICTBT(-1)	0.8345***	0.0402	20.7585	0.0000
D(VIX)	-0.0037***	0.0011	-3.3060	0.0011
D(MOVE)	-0.0001	0.0003	-0.4299	0.6678
D(IP)	0.0010	0.0011	0.9768	0.3299
<i>Adj R</i>²	71.8%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Tables 5.5.6 and 5.5.7 show the robustness checks results for Canadian 3-month Treasury and total government T-bill. There are thresholds detected with significantly negative coefficients of stock market and significantly positive lagged dependent variables in both series. Among additional controls, D(VIX) is significantly negative, indicating that when the stock markets are expected to be riskier in the future, the short-term bond returns will fall. These results further confirm the robustness of previous findings.

Table 5.5.8: Threshold Regression Results on SPFCGBT for Robustness (Sample: 1998M02–2023M09) with additional controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0062	0.0211	0.2927	0.7700
Inflation	-0.0837*	0.0490	-1.7093	0.0884
C	0.4971***	0.1259	3.9488	0.0001
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0051	0.0186	0.2718	0.7860
D(MOVE)	-0.0163***	0.0049	-3.3306	0.0010
D(IP)	0.0108	0.0141	0.7668	0.4438
<i>Adj R²</i>	4.0%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.9: Threshold Regression Results on SPFCAGG for Robustness (Sample: 1998M02–2023M09) with additional controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0159	0.0196	0.8109	0.4180
Inflation	-0.0801*	0.0457	-1.7539	0.0805
C	0.1368	0.1174	1.1652	0.2448
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0059	0.0174	0.3369	0.7364
D(MOVE)	-0.0159***	0.0046	-3.4694	0.0006
D(IP)	0.0116	0.0132	0.8795	0.3798
<i>Adj R²</i>	5.1%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Tables 5.5.8 and 5.5.9 show the robustness checks results for Canadian government bond and aggregate bond. For both series, no significant threshold is detected, suggesting that the regime-dependent dynamics observed in other segments are largely explained by inflation and macro-financial uncertainty. Inflation remains marginally negative and significant at the 10% level in both models, while stock returns become insignificant after controls are added. Among the additional controls, only D(MOVE) is consistently significant and negative, reflecting the sensitivity of bond returns to interest rate volatility. These results confirm the robustness of the earlier results.

Table 5.5.10: Threshold Regression Results on SPFCG1ST for Robustness (Sample: 1998M02–2023M09) with additional controls — Canada

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.50%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0286**	0.0118	-2.4237	0.0160
Inflation	0.0099	0.0570	0.1734	0.8625
C	0.2532***	0.0950	2.6647	0.0081
<i>Inflation ≥ 2.50%</i>				
<i>(High Regime)</i>				
Stock market	-0.0059	0.0135	-0.4390	0.6609
Inflation	-0.1804***	0.0427	-4.2200	0.0000
C	0.9728***	0.1732	5.6165	0.0000
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0000	0.0089	0.0034	0.9973
D(MOVE)	-0.0065***	0.0024	-2.7674	0.0060
D(IP)	-0.0026	0.0067	-0.3861	0.6997
<i>Adj R²</i>	7.5%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.10 reports the robustness checks results for Canadian 1-5 years government bond. The model continues to detect a threshold at 2.5% inflation, and the key findings remain robust. In the low-inflation regime, the stock-bond correlation remains significantly negative, while inflation is insignificant. In the high-inflation regime, the stock-bond correlation becomes insignificant, while inflation retains a strong and significant negative effect. Among the additional controls, only D(MOVE) is significant and negative, reflecting the sensitivity of medium-term bond returns to interest rate volatility. These results confirm the robustness of the earlier findings, highlighting the regime-dependent nature of 1–5 year bond returns and their vulnerability to high inflation and interest rate uncertainty.

5.5.4 Eurozone

Table 5.5.11: Threshold Regression Results on EMTT3UC (Sample: 2003M10–2025M03) for Robustness with additional controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold detected				
Stock market	0.0001	0.0006	0.1662	0.8681
Inflation	-0.0025**	0.0010	-2.3977	0.0172
C	0.0068**	0.0030	2.2906	0.0228
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
EMTT3UC(-1)	0.9807***	0.0150	65.2580	0.0000
D(VIX)	0.0002	0.0005	0.3438	0.7313
D(MOVE)	0.0001	0.0001	0.5182	0.6048
D(IP)	-0.0004	0.0010	-0.3812	0.7034
<i>Adj R²</i>	94.6%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.11 reports the robustness checks results for 3-month Eurozone government bill returns. The regression results remain consistent with the earlier findings. The threshold structure does not exist, and the inflation variable continues to exhibit a significant and negative effect on 3-month bond returns. Stock returns remain statistically insignificant, and the additional controls themselves are not significant. This suggests that the negative impact of inflation is robust to the inclusion of macro-financial uncertainty controls, while the stock-bond relationship remains weak.

Table 5.5.12: Threshold Regression Results on EXEAS for Robustness (Sample: 2007M03–2025M03) with additional controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 2.10%</i>				
<i>(Low Regime)</i>				
Stock market	0.0115	0.0094	1.2155	0.2256
Inflation	-0.0357	0.0531	-0.6727	0.5019
C	0.1069*	0.0623	1.7151	0.0878
<i>Inflation ≥ 2.10%</i>				
<i>(High Regime)</i>				
Stock market	-0.0025	0.0101	-0.2501	0.8028
Inflation	-0.0897***	0.0205	-4.3862	0.0000
C	0.4866***	0.0931	5.2284	0.0000
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0060	0.0077	0.7858	0.4329
D(MOVE)	-0.0020	0.0021	-0.9614	0.3375
D(IP)	-0.0145	0.0145	-1.0006	0.3182
<i>Adj R²</i>	7.5%			

Significance is denoted by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 5.5.12 reports the robustness checks results for 1–3 years Eurozone government bond returns. The model continues to identify a significant threshold at 2.1%, and the high-inflation regime remains associated with a strong and significant negative effect of inflation on bond returns. Stock returns remain insignificant in both regimes. The additional controls do not attain significance or alter the main findings.

Table 5.5.13: Threshold Regression Results on EXEBS for Robustness (Sample: 2007M03–2025M03) with additional controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0122	0.0148	0.8252	0.4102
Inflation	-0.0778**	0.0262	-2.9714	0.0033
C	0.3223***	0.0790	4.0827	0.0001
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0108	0.0145	0.7475	0.4556
D(MOVE)	-0.0065	0.0039	-1.6513	0.1002
D(IP)	-0.0098	0.0273	-0.3582	0.7205
<i>Adj R²</i>	3.95%			

Significance is denoted by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 5.5.13 reports the robustness checks results for 3–5 years Eurozone government bond returns. It is worth noting that unlike the previous results, the model no longer identifies a significant threshold, and stock-bond correlation becomes insignificant. This suggests that the regime-dependent dynamics observed in the baseline results are at least partly driven by broader macro-financial uncertainty rather than inflation solely. These findings confirm the robustness of the baseline results in the sense that they reveal the underlying mechanism that the inflation threshold effect captures the impact of risk and uncertainty that coincide with high-inflation periods in the samples selected.

Table 5.5.14: Threshold Regression Results on EXEC5 for Robustness(Sample: 2004M09–2025M03) with additional controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0151	0.0203	0.7438	0.4577
Inflation	-0.1188***	0.0364	-3.2622	0.0013
C	0.4720***	0.1069	4.4143	0.0000
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0078	0.0201	0.3878	0.6985
D(MOVE)	-0.0112*	0.0054	-2.0678	0.0397
D(IP)	0.0064	0.0375	0.1712	0.8642
<i>Adj R²</i>	5.2%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.14 reports the robustness checks results for 5–7 years Eurozone government bond returns. The results show that no significant threshold is detected, suggesting that the regime-dependent dynamics observed earlier are largely explained by inflation and macro-financial uncertainty. Inflation remains significantly negative, confirming its role in driving bond returns even after controlling for these additional factors. Among the controls, only D(MOVE) is significant, indicating that interest rate uncertainty also contributes to bond return variation, while stock returns remain insignificant throughout. These findings reinforce the robustness of the results and highlight the role of inflation and macro-financial risks.

Table 5.5.15: Threshold Regression Results on EXED5 for Robustness (Sample: 2008M01–2025M03) with additional controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.40%</i>				
<i>(Low Regime)</i>				
Stock market	0.0099	0.0293	0.3387	0.7352
Inflation	-0.0293	0.1181	-0.2480	0.8044
C	0.4345**	0.2006	2.1657	0.0315
<i>Inflation ≥ 3.40%</i>				
<i>(High Regime)</i>				
Stock market	0.2348***	0.0597	3.9367	0.0001
Inflation	-0.3524***	0.1311	-2.6872	0.0078
C	1.8006**	0.8969	2.0076	0.0460
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0051	0.0276	0.1853	0.8532
D(MOVE)	-0.0099	0.0078	-1.2753	0.2037
D(IP)	0.0233	0.0517	0.4499	0.6533
<i>Adj R²</i>	13.1%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.15 reports the robustness checks results for 7–10 years Eurozone government bond returns. The model continues to detect a significant threshold at 3.40%, confirming the regime-dependent dynamics. In the low-inflation regime, stock returns and inflation remain insignificant. In the high-inflation regime, stock returns exhibit a strong and significant positive association with bond returns, and inflation has a significant negative effect. The additional controls are not significant and do not materially affect the main results. These findings confirm the robustness of the baseline results and underscore the influence of high inflation on stock-bond comovement and bond performance.

Table 5.5.16: Threshold Regression Results on EXEES for Robustness(Sample: 2007M03–2025M03) with additional controls — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0795**	0.0374	2.1248	0.0348
Inflation	-0.2238***	0.0661	-3.3873	0.0008
C	0.7278***	0.1993	3.6516	0.0003
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0136	0.0366	0.3724	0.7100
D(MOVE)	-0.0179*	0.0100	-1.8010	0.0731
D(IP)	0.0284	0.0690	0.4111	0.6814
<i>Adj R²</i>	9.5%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.16 shows the robustness checks results for 10–15 years Eurozone government bond returns. Unlike the previous results, no significant threshold is detected for 10–15 year bond returns after adding additional controls, suggesting that the regime-dependent dynamics observed earlier are largely explained by inflation and macro-financial uncertainty. Nevertheless, stock returns remain significantly and positively associated with bond returns, while inflation continues to exert a significant negative effect. Among the controls, only D(MOVE) is marginally significant, reflecting the influence of interest rate uncertainty. However, although no significant threshold is detected after including additional control variables, this still serves as a robustness check because it demonstrates that the key findings remain consistent even when accounting for macro-financial uncertainty.

Table 5.5.17: Threshold Regression Results on EXIEF5 for Robustness(Sample: 2004M02–2025M03) — Eurozone

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	0.0571	0.0551	1.0377	0.3004
Inflation	-0.3655***	0.1002	-3.6478	0.0003
C	1.0142***	0.2924	3.4681	0.0006
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	-0.0070	0.0549	-0.1274	0.8988
D(MOVE)	-0.0235	0.0148	-1.5817	0.1150
D(IP)	0.0740	0.1027	0.7204	0.4720
<i>Adj R²</i>	6.7%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.17 reports the robustness checks results for 15-year and above Eurozone government bond returns. The results continuously show that no significant threshold is detected, suggesting that the regime-dependent dynamics observed earlier are largely accounted for by inflation and macro-financial uncertainty. Inflation continues to have a significant and negative effect on bond returns, confirming the robustness of the main findings. The stock-bond correlation becomes insignificant, while the controls themselves remain insignificant. These results reinforce the conclusion that inflation is a key driver of long-term bond returns, even when controlling for additional sources of risk.

5.5.5 UK

Table 5.5.18: Threshold Regression Results on FTRFBGL for Robustness (Sample: 2004M04–2025M03) with additional controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 4.00%</i>				
<i>(Low Regime)</i>				
Stock market	-0.002	0.0142	-0.1441	0.8855
Inflation	-0.0055	0.0426	-0.1282	0.8981
C	0.2009**	0.0949	2.1171	0.0353
<i>Inflation ≥ 4.00%</i>				
<i>(High Regime)</i>				
Stock market	0.015	0.0243	0.6188	0.5366
Inflation	-0.1361***	0.0362	-3.761	0.0002
C	1.1243***	0.2552	4.4063	0.0000
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.005	0.01	0.4971	0.6196
D(MOVE)	-0.0028	0.0028	-1.0174	0.31
D(IP)	-0.0045	0.0205	-0.2213	0.8251
<i>Adj R²</i>	3.1%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.19: Threshold Regression Results on FTRFBGM for Robustness (Sample: 2004M04–2025M03) with additional controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 4.00%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0465	0.0429	-1.0841	0.2794
Inflation	-0.0488	0.1287	-0.3796	0.7048
C	0.3866	0.2865	1.3494	0.1785
<i>Inflation ≥ 4.00%</i>				
<i>(High Regime)</i>				
Stock market	0.1114	0.0733	1.5194	0.13
Inflation	-0.4276***	0.1093	-3.9132	0.0001
C	3.1059***	0.7704	4.0316	0.0001
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	-0.0013	0.0304	-0.0413	0.9671
D(MOVE)	-0.015*	0.0084	-1.777	0.0768
D(IP)	0.0117	0.062	0.1891	0.8502
<i>Adj R²</i>	6.0%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.20: Threshold Regression Results on FTRFBGM1 for Robustness (Sample: 1999M08–2025M03) with additional controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0434	0.0301	-1.4446	0.1496
Inflation	-0.089	0.1028	-0.8661	0.3871
C	0.4804**	0.2083	2.3063	0.0218
<i>Inflation ≥ 3.80%</i>				
<i>(High Regime)</i>				
Stock market	0.1325**	0.0599	2.2124	0.0277
Inflation	-0.3807***	0.0913	-4.1713	0.0000
C	2.818***	0.6348	4.4391	0.0000
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0188	0.0242	0.7788	0.4367
D(MOVE)	-0.0141**	0.0065	-2.1779	0.0302
D(IP)	-0.0064	0.0493	-0.1291	0.8974
<i>Adj R²</i>	7.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.21: Threshold Regression Results on FTREBGM2 for Robustness (Sample: 1999M08–2025M03) with additional controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0552	0.0421	-1.3105	0.191
Inflation	-0.119	0.1438	-0.8275	0.4086
C	0.5596*	0.2915	1.9198	0.0558
<i>Inflation ≥ 3.80%</i>				
<i>(High Regime)</i>				
Stock market	0.1896**	0.0838	2.2632	0.0243
Inflation	-0.5333***	0.1277	-4.176	0.0000
C	3.7693***	0.8882	4.2435	0.0000
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0255	0.0338	0.7533	0.4519
D(MOVE)	-0.0217**	0.009	-2.4006	0.017
D(IP)	-0.0051	0.0689	-0.0733	0.9416
<i>Adj R²</i>	<i>7.7%</i>			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.22: Threshold Regression Results on FTREBGH for Robustness (Sample: 1999M08–2025M03) with additional controls — UK

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 3.80%</i>				
<i>(Low Regime)</i>				
Stock market	-0.0374	0.0645	-0.5802	0.5622
Inflation	-0.2311	0.2204	-1.0485	0.2953
C	0.7926*	0.4467	1.7744	0.077
<i>Inflation ≥ 3.80%</i>				
<i>(High Regime)</i>				
Stock market	0.2896**	0.1284	2.2557	0.0248
Inflation	-0.8585***	0.1957	-4.3862	0.0000
C	5.3496***	1.3612	3.93	0.0001
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0467	0.0518	0.9013	0.3681
D(MOVE)	-0.0247*	0.0138	-1.7838	0.0755
D(IP)	-0.1067	0.1056	-1.0098	0.3134
<i>Adj R²</i>	7.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.18-5.5.22 report the robustness checks results for UK's 0–5 years, 5–15 years, 5–10 years, 10–15 years and 15-year and above government bond returns. The overall results continue to exhibit a clear threshold effect, with the stock–bond relationship remaining stronger in the high-inflation regime. Compared to the baseline regression, however, the significance of the high-inflation effect diminishes, with the 5–15 years segment becoming insignificant in the high-inflation regime, and the 5–10 years segment losing its significance in the low-inflation regime. The results for other maturities remain largely unchanged, and the general pattern—stronger stock-bond comovement under high inflation and weaker under low inflation—persists. The change of significance appears to reflect the fact that the added control variables capture aspects of macro-financial uncertainty that are closely tied to high-inflation periods. These factors tend to rise alongside inflation, absorbing part of the effect previously attributed solely to high inflation. The findings remain robust after accounting for macro-financial uncertainty, although the significance of some results in the high-inflation regime weakens, suggesting that part of the stock-bond comovement under high inflation is

linked to heightened risk and uncertainty.

5.5.6 Switzerland

Table 5.5.23: Threshold Regression Results on SPFISWUT for Robustness(Sample: 2004M02–2025M03) with additional — Switzerland

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0045	0.0304	-0.1482	0.8823
Inflation	-0.0141	0.0805	-0.175	0.8612
C	0.1853*	0.0996	1.8613	0.0639
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	-0.0004	0.0217	-0.0171	0.9864
D(MOVE)	-0.0002	0.0066	-0.0251	0.98
D(IP)	-0.0984***	0.0347	-2.8365	0.0049
<i>Adj R²</i>	1.3%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.23 reports the robustness checks results for Switzerland government bond returns. The results continuously show that there is no stock–bond correlation in Switzerland, which is consistent with previous findings.

5.5.7 China

Table 5.5.24: Threshold Regression Results for Returns on SPBCN1BT (Sample: 2007M01–2025M03) with additional controls — China

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
No threshold selected				
Stock market	-0.0449***	0.01211	-3.7081	0.0003
Inflation	-0.0094	0.04324	-0.2183	0.8274
C	0.3825***	0.1301	2.9409	0.0036
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
D(VIX)	0.0056	0.0176	0.3178	0.751
D(MOVE)	0.004	0.0059	0.677	0.4991
D(IP)	0.0000	0.0000	-0.0613	0.9512
<i>Adj R²</i>	5.4%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.24 reports the robustness checks results for China’s 10-year government bond returns. After adding additional controls, the regression still shows a significant negative relationship between stock returns and China’s 10-year government bond returns, indicating that this result is robust. Inflation and the added control variables are not statistically significant, suggesting they have little direct explanatory power for bond returns and do not alter the main stock–bond relationship. The negative stock–bond correlation remains strong and significant, supporting the robustness of the earlier findings.

Table 5.5.25: Threshold Regression Results on SPBCNGOT for Robustness (Sample: 2007M01–2025M03) with additional controls — China

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Statistic</i>	<i>p-Value</i>
<i>Inflation < 1.5%</i>				
Stock market	0.0163	0.0109	1.4896	0.1379
Inflation	0.2540***	0.0764	3.3236	0.0011
C	0.2022***	0.0736	2.7467	0.0066
<i>1.5% ≤ Inflation < 2.5%</i>				
Stock market	-0.0081	0.0094	-0.8635	0.3889
Inflation	0.5983***	0.2132	2.8067	0.0055
C	-0.9940**	0.4206	-2.3630	0.0191
<i>2.5% ≤ Inflation < 4.30%</i>				
Stock market	-0.0277**	0.0108	-2.5688	0.0109
Inflation	0.6222***	0.1719	3.6191	0.0004
C	-1.7294***	0.5177	-3.3404	0.0010
<i>Inflation ≥ 4.30%</i>				
Stock market	0.0175*	0.0099	1.7687	0.0785
Inflation	0.1087	0.0762	1.4270	0.1551
C	-0.5622	0.4557	-1.2336	0.2188
<i>Non-Threshold</i>				
<i>Variables as Controls</i>				
SPBCNGOT(-1)	0.4829***	0.0592	8.1645	0.0000
D(VIX)	-0.0110	0.0076	-1.4425	0.1507
D(MOVE)	0.0043*	0.0025	1.7139	0.0881
D(IP)	0.0000	0.0000	1.3041	0.1937
<i>Adj R²</i>	33.6%			

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.25 reports the robustness checks results for China's government bond returns. The model continues to detect three threshold values and four regimes, which is consistent with the previous results. In the first two regimes, the stock–bond correlation does not exist. In the regime where $2.5\% \leq \text{Inflation} < 4.30\%$, the stock returns are negatively correlated with the bond returns, and this relationship disappears when inflation is above 4.30%. Among the additional controls, D(MOVE) is the only significant and positive predictor, suggesting

that when investors expect higher risk and volatility in the bond market (as reflected by an increase in the MOVE index), they tend to exit the market. This selling pressure lowers bond prices, which, by construction, raises yields and may increase bond returns. These findings further support the robustness of early ones.

In all, we find some support that there exist inflation thresholds that separate the stock-bond relationship into the high- and low-inflation regimes, and that stock and bonds are positively related under the high-inflation regime. The empirical support on a per-country basis leads us to extend the empirical analysis by considering the panel dimensions of our data, in what follows, via a panel analysis. This enables examining the inflation regime-dependent channel of the stock-bond relationship in a panel framework.

5.5.8 Panel Data Analysis

In the last section, we have so far looked at a time series analysis by country. Given some of the encouraging support of our main result on a country-by-country basis, we now take advantage of the cross-sectional and time-series structure of the data and perform a panel analysis. To this end, we examine the inflation regime channel of the stock-bond relationship for the markets across time. The Hansen (1999) panel threshold framework we employed has asymptotic properties that benefit from high time dimensions. For instance, the long time series renders incidental parameter bias negligible and enhances precision in estimating both the threshold parameter and regime-specific relationships. It also allows robust inference and facilitates specification testing. Thus, the long panel framework satisfies the asymptotic validity conditions fundamental to reliable threshold estimation.

We begin with estimating the regressions using panel techniques on the empirical specification in equation (6). In the table below, we present the panel framework. The estimated coefficients continue to advance the finding that there is an inflation regime channel that explains the stock-bond relationship and that in a high inflation regime, the stock and bond market are strongly related positively.

Table 5.5.26: Threshold Regression Results for Panel Data with additional controls

	Panel A	Panel B	Panel C
	3-month	5-year	10-year
Threshold Value $\gamma\%$	7.4000 (0.3000)	6.9000*** (0.0000)	6.2000*** (0.0000)
<i>Inflation < $\gamma\%$</i> <i>(Low Regime)</i>			
Stock Market	-0.0237*** (0.0000)	-0.0536*** (0.0000)	-0.0511*** (0.0000)
Inflation	0.0151 (0.2040)	0.0060 (0.5900)	0.0250 (0.4540)
<i>Inflation $\geq \gamma\%$</i> <i>(High Regime)</i>			
Stock Market	0.0674** (0.0150)	0.2082*** (0.0000)	0.0556* (0.0740)
Inflation	-0.0810*** (0.0000)	-0.1251*** (0.0000)	-0.1450*** (0.0000)
C	0.0855** (0.0130)	0.2325*** (0.0010)	0.2776*** (0.0000)
<i>Non-Threshold</i> <i>Variables as Controls</i>			
D(VIX)	-0.0033 (0.5550)	-0.0102 (0.3250)	-0.0085 (0.3760)
D(MOVE)	-0.0022 (0.1580)	-0.0082*** (0.0030)	-0.0077** (0.0120)
D(IP)	0.0008 (0.9000)	-0.0013 (0.9120)	0.0000 (0.7330)
Within R^2	7.35%	14.44%	8.07%

Significance is denoted by * $p < .10$; ** $p < .05$; *** $p < .01$.

Table 5.5.26 reports the robustness checks results for panel data. Same as the time-series analysis, we add D(VIX), D(MOVE), and D(IP) to further check the robustness of panel threshold regression results. The estimated threshold values remain consistent with the baseline results (7.400, 6.900, and 6.200 for the 3-month, 5-year, and 10-year horizons,

respectively), all within reasonable confidence intervals. The threshold effect tests indicate that the threshold is not significant at the 3-month horizon, but remains strongly significant at the 5-year and 10-year horizons (both $p < 0.01$), suggesting that the threshold effect is more salient over longer maturities. The main coefficients of interest also remain stable: the interaction terms for stock returns are negative and significant below the threshold, and positive and significant above the threshold, with the 10-year horizon showing marginal significance above the threshold. Inflation retains its negative and significant effect in the high-inflation regime, consistent with the baseline findings. Among the control variables, $D(\text{MOVE})$ exhibits a significant negative effect at the 5-year and 10-year horizons. This suggests that bond returns at longer maturities are more sensitive to increases in market uncertainty, as measured by bond market implied volatility. When $D(\text{MOVE})$ rises, signaling heightened risk aversion and greater expected bond price fluctuations, longer-term bond returns decline. This finding aligns with the notion of term premia, whereby longer-term bonds carry higher risk and respond more strongly to changes in uncertainty. Overall, the inclusion of additional controls does not alter the main conclusions. The threshold effect remains robust and significant at medium and long horizons, while the short-term result is directionally consistent but statistically insignificant.

6 Conclusion

This paper develops a continuous-time model that provides a rigorous foundation for analyzing regime-dependent comovement between asset classes, with clear implications for dynamic asset allocation. We show that inflation volatility and investor beliefs about regime transitions fundamentally shape the risk–return tradeoff in financial markets, especially the correlation structure between stocks and bonds.

A foundational assumption in modern portfolio theory is that stocks and nominal bonds are negatively correlated, offering diversification benefits. However, a growing empirical literature has documented that this correlation is neither stable nor universal. Periods of high and volatile inflation—such as the 1970s or post-pandemic monetary tightening—often coincide with positive stock–bond correlations, undermining traditional allocation strategies.

We extend this literature by introducing a continuous-time framework in which inflation follows a regime-switching Ornstein–Uhlenbeck process, and both stock and bond returns exhibit state-dependent sensitivity to inflation shocks. This setup enables us to explicitly model how inflation volatility and the probability of regime persistence or transition affect asset return correlations. Importantly, the model yields tractable, closed-form expressions for instantaneous correlation and its comparative statics, allowing us to state and prove a

series of economically interpretable propositions.

This paper makes two central contributions. First, we develop a tractable continuous-time model in which inflation regimes endogenously drive changes in the covariance structure of asset returns. We derive seven propositions that characterize how stock–bond correlation responds to inflation variance, asymmetric inflation sensitivities, and regime transition probabilities—including nonlinear amplification effects. Second, we empirically evaluate these predictions using data from eight major economies, documenting robust evidence that the correlation between equity and bond returns is strongly regime-dependent and rises with both inflation volatility and the perceived probability of persistent inflation.

7 Reference

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8 Appendix Proofs

Proposition 1: Regime Dependence of Correlation

Proof: This follows directly from the correlation formula. Because $\text{Var}(\pi_t^H) > \text{Var}(\pi_t^L)$ and $|\alpha_S^H| > |\alpha_S^L|$, $|\alpha_B^H| > |\alpha_B^L|$, the numerator grows faster than the denominator. Hence, $\rho_{SB}^H > \rho_{SB}^L$.

More formally, we model stock and bond returns as driven by inflation shocks with regime-dependent sensitivities. Let inflation π_t follow an exogenous process with two regimes (low and high), and let both asset returns be affine functions of inflation shocks as noted in the main model:

$$\begin{aligned} dR_t^S &= \mu_S dt + \alpha_S(\theta) d\pi_t + \sigma_S dW_t^S, \\ dR_t^B &= \mu_B dt + \alpha_B(\theta) d\pi_t + \sigma_B dW_t^B, \end{aligned}$$

where:

- $\alpha_S(\theta), \alpha_B(\theta)$ are inflation betas of stocks and bonds, varying by regime $\theta \in \{L, H\}$,
- dW_t^S, dW_t^B are Brownian motions orthogonal to $d\pi_t$,
- π_t evolves according to an Ornstein–Uhlenbeck process:
 $d\pi_t = \kappa(\bar{\pi}(\theta) - \pi_t)dt + \sigma_\pi dZ_t$, with $dZ_t \sim \mathcal{N}(0, dt)$, where $\bar{\pi}(\theta)$ is the regime-dependent long-run mean inflation level, $\kappa > 0$ is the speed of mean reversion, and $dZ_t \sim \mathcal{N}(0, dt)$.

The instantaneous covariance between stock and bond returns is:

$$\begin{aligned} \text{Cov}(dR_t^S, dR_t^B) &= \text{Cov}(\alpha_S(\theta)d\pi_t + \sigma_S dW_t^S, \alpha_B(\theta)d\pi_t + \sigma_B dW_t^B) \\ &= \alpha_S(\theta)\alpha_B(\theta)\text{Var}(d\pi_t) \\ &= \alpha_S(\theta)\alpha_B(\theta)\sigma_\pi^2 dt. \end{aligned}$$

The instantaneous variances are:

$$\begin{aligned} \text{Var}(dR_t^S) &= \alpha_S^2(\theta)\sigma_\pi^2 dt + \sigma_S^2 dt, \\ \text{Var}(dR_t^B) &= \alpha_B^2(\theta)\sigma_\pi^2 dt + \sigma_B^2 dt. \end{aligned}$$

Hence, the instantaneous correlation is:

$$\rho_{SB}(\theta) = \frac{\alpha_S(\theta)\alpha_B(\theta)\sigma_\pi^2}{\sqrt{\alpha_S^2(\theta)\sigma_\pi^2 + \sigma_S^2} \cdot \sqrt{\alpha_B^2(\theta)\sigma_\pi^2 + \sigma_B^2}}.$$

Define $\rho_{SB}^\theta := \rho_{SB}(\theta)$ for $\theta \in \{L, H\}$. We now analyze:

$$|\rho_{SB}^H| > |\rho_{SB}^L|.$$

This inequality holds when the numerator increases more rapidly than the denominator. Suppose:

$$|\alpha_S(H)| > |\alpha_S(L)| \quad \text{and} \quad |\alpha_B(H)| > |\alpha_B(L)|.$$

Then:

- Covariance increases in magnitude since $\alpha_S(H)\alpha_B(H)\sigma_\pi^2 > \alpha_S(L)\alpha_B(L)\sigma_\pi^2$,
- The denominator increases more slowly due to the sublinearity of the square root.

Hence,

$$|\rho_{SB}^H| > |\rho_{SB}^L|.$$

Positive and Negative Correlation Domains

- If $\alpha_S, \alpha_B > 0$, then $\rho_{SB}^\theta > 0$ and $\rho_{SB}^H > \rho_{SB}^L$.
- If $\alpha_S > 0, \alpha_B < 0$ or vice versa, then $\rho_{SB}^\theta < 0$ and $\rho_{SB}^H < \rho_{SB}^L$, implying:

$$|\rho_{SB}^H| > |\rho_{SB}^L|.$$

- Thus, in the negative domain ($\rho_{SB}^\theta \in [-1, 0)$), we still have stronger magnitude in high inflation:

$$-\rho_{SB}^H < -\rho_{SB}^L \quad \text{and} \quad |\rho_{SB}^H| > |\rho_{SB}^L|.$$

We have used the fact that the instantaneous covariance between dR_t^S and dR_t^B :

$$\text{Cov}(dR_t^S, dR_t^B) = \text{Cov}(\alpha_S(\theta)d\pi_t + \sigma_S dW_t^S, \alpha_B(\theta)d\pi_t + \sigma_B dW_t^B).$$

Expanding using bilinearity of covariance:

$$\begin{aligned} \text{Cov}(dR_t^S, dR_t^B) &= \text{Cov}(\alpha_S(\theta)d\pi_t, \alpha_B(\theta)d\pi_t) \\ &\quad + \text{Cov}(\alpha_S(\theta)d\pi_t, \sigma_B dW_t^B) \\ &\quad + \text{Cov}(\sigma_S dW_t^S, \alpha_B(\theta)d\pi_t) \\ &\quad + \text{Cov}(\sigma_S dW_t^S, \sigma_B dW_t^B). \end{aligned}$$

Each term is evaluated as follows:

- $\text{Cov}(\alpha_S(\theta)d\pi_t, \alpha_B(\theta)d\pi_t) = \alpha_S(\theta)\alpha_B(\theta)\text{Var}(d\pi_t),$
- $\text{Cov}(\alpha_S(\theta)d\pi_t, \sigma_B dW_t^B) = 0$ (independence),
- $\text{Cov}(\sigma_S dW_t^S, \alpha_B(\theta)d\pi_t) = 0$ (independence),
- $\text{Cov}(\sigma_S dW_t^S, \sigma_B dW_t^B) = 0$ (independence).

Thus:

$$\text{Cov}(dR_t^S, dR_t^B) = \alpha_S(\theta)\alpha_B(\theta)\text{Var}(d\pi_t).$$

Since $d\pi_t = \kappa(\bar{\pi} - \pi_t)dt + \sigma_\pi dZ_t$, the diffusion component is $\sigma_\pi dZ_t$, and hence:

$$\text{Var}(d\pi_t) = \sigma_\pi^2 dt.$$

Therefore, the final expression is:

$$\text{Cov}(dR_t^S, dR_t^B) = \alpha_S(\theta)\alpha_B(\theta)\sigma_\pi^2 dt.$$

Also, for the variance of the stock and bond returns, we have used the fact that

*Variance of Stock Return

$$\begin{aligned}\text{Var}(dR_t^S) &= \text{Var}(\alpha_S(\theta)d\pi_t + \sigma_S dW_t^S) \\ &= \text{Var}(\alpha_S(\theta)d\pi_t) + \text{Var}(\sigma_S dW_t^S) + 2\text{Cov}(\alpha_S(\theta)d\pi_t, \sigma_S dW_t^S).\end{aligned}$$

Each component is, by Ito Lemma:

$$\begin{aligned}\text{Var}(\alpha_S(\theta)d\pi_t) &= \alpha_S^2(\theta) \cdot \text{Var}(d\pi_t) = \alpha_S^2(\theta) \cdot \sigma_\pi^2 dt, \\ \text{Var}(\sigma_S dW_t^S) &= \sigma_S^2 \cdot \text{Var}(dW_t^S) = \sigma_S^2 dt, \\ \text{Cov}(\alpha_S(\theta)d\pi_t, \sigma_S dW_t^S) &= 0 \quad (\text{independence}).\end{aligned}$$

Thus, the total variance is:

$$\text{Var}(dR_t^S) = \alpha_S^2(\theta)\sigma_\pi^2 dt + \sigma_S^2 dt.$$

Also, *Variance of Bond Return

$$\begin{aligned}
\text{Var}(dR_t^B) &= \text{Var}(\alpha_B(\theta)d\pi_t + \sigma_B dW_t^B) \\
&= \text{Var}(\alpha_B(\theta)d\pi_t) + \text{Var}(\sigma_B dW_t^B) + 2 \text{Cov}(\alpha_B(\theta)d\pi_t, \sigma_B dW_t^B).
\end{aligned}$$

Evaluate each component, by Ito Lemma:

$$\begin{aligned}
\text{Var}(\alpha_B(\theta)d\pi_t) &= \alpha_B^2(\theta) \cdot \text{Var}(d\pi_t) = \alpha_B^2(\theta) \cdot \sigma_\pi^2 dt, \\
\text{Var}(\sigma_B dW_t^B) &= \sigma_B^2 \cdot \text{Var}(dW_t^B) = \sigma_B^2 dt, \\
\text{Cov}(\alpha_B(\theta)d\pi_t, \sigma_B dW_t^B) &= 0 \quad (\text{independence}).
\end{aligned}$$

Hence, the bond return variance is:

$$\text{Var}(dR_t^B) = \alpha_B^2(\theta)\sigma_\pi^2 dt + \sigma_B^2 dt.$$

These expressions are then used to compute the correlation between stock and bond returns in each regime $\theta \in \{L, H\}$ as follows: instantaneous correlation between stock and bond returns in regime $\theta \in \{L, H\}$ is defined as:

$$\rho_{SB}(\theta) = \frac{\text{Cov}(dR_t^S, dR_t^B)}{\sqrt{\text{Var}(dR_t^S)} \cdot \sqrt{\text{Var}(dR_t^B)}}.$$

From the previous derivations, we have:

$$\begin{aligned}
\text{Cov}(dR_t^S, dR_t^B) &= \text{Cov}(\alpha_S(\theta)d\pi_t + \sigma_S dW_t^S, \alpha_B(\theta)d\pi_t + \sigma_B dW_t^B) \\
&= \alpha_S(\theta)\alpha_B(\theta) \cdot \text{Var}(d\pi_t) + 0 + 0 + 0 \\
&= \alpha_S(\theta)\alpha_B(\theta) \cdot \sigma_\pi^2 dt.
\end{aligned}$$

Also, we have:

$$\begin{aligned}
\text{Var}(dR_t^S) &= \alpha_S^2(\theta)\sigma_\pi^2 dt + \sigma_S^2 dt = dt \cdot (\alpha_S^2(\theta)\sigma_\pi^2 + \sigma_S^2), \\
\text{Var}(dR_t^B) &= \alpha_B^2(\theta)\sigma_\pi^2 dt + \sigma_B^2 dt = dt \cdot (\alpha_B^2(\theta)\sigma_\pi^2 + \sigma_B^2).
\end{aligned}$$

Thus, the correlation becomes:

$$\begin{aligned}\rho_{SB}(\theta) &= \frac{\alpha_S(\theta)\alpha_B(\theta)\sigma_\pi^2 dt}{\sqrt{(\alpha_S^2(\theta)\sigma_\pi^2 + \sigma_S^2) dt} \cdot \sqrt{(\alpha_B^2(\theta)\sigma_\pi^2 + \sigma_B^2) dt}} \\ &= \frac{\alpha_S(\theta)\alpha_B(\theta)\sigma_\pi^2}{\sqrt{\alpha_S^2(\theta)\sigma_\pi^2 + \sigma_S^2} \cdot \sqrt{\alpha_B^2(\theta)\sigma_\pi^2 + \sigma_B^2}}.\end{aligned}$$

Proposition 2: Correlation Increases with Inflation Variance

The instantaneous correlation between stock and bond returns, $\rho_{SB}(\theta)$, increases in magnitude with the variance of inflation shocks σ_π^2 , for any fixed values of inflation betas $\alpha_S(\theta), \alpha_B(\theta)$ and idiosyncratic volatilities σ_S, σ_B .

Proof: Recall that the instantaneous correlation between stock and bond returns in regime $\theta \in \{L, H\}$ is given by:

$$\rho_{SB}(\theta) = \frac{\alpha_S(\theta)\alpha_B(\theta)\sigma_\pi^2}{\sqrt{\alpha_S^2(\theta)\sigma_\pi^2 + \sigma_S^2} \cdot \sqrt{\alpha_B^2(\theta)\sigma_\pi^2 + \sigma_B^2}}. \quad (7)$$

Let us define:

$$A := \alpha_S(\theta), \quad B := \alpha_B(\theta), \quad s := \sigma_S, \quad b := \sigma_B.$$

Then, Equation (7) becomes:

$$\rho_{SB}(\theta) = \frac{AB \cdot \sigma_\pi^2}{\sqrt{A^2\sigma_\pi^2 + s^2} \cdot \sqrt{B^2\sigma_\pi^2 + b^2}}. \quad (8)$$

We now investigate how $\rho_{SB}(\theta)$ changes with respect to σ_π^2 .

Let $x := \sigma_\pi^2$, and define:

$$f(x) := \rho_{SB}(\theta) = \frac{AB \cdot x}{\sqrt{A^2x + s^2} \cdot \sqrt{B^2x + b^2}}.$$

Objective: Show that $\frac{d}{dx}|f(x)| > 0$.

Let us proceed by computing the derivative $f'(x)$. First, denote the denominator as:

$$D(x) = \sqrt{A^2x + s^2} \cdot \sqrt{B^2x + b^2}.$$

Then,

$$f(x) = \frac{AB \cdot x}{D(x)}.$$

Use the quotient rule:

$$f'(x) = \frac{AB \cdot D(x) - AB \cdot x \cdot D'(x)}{D(x)^2}.$$

So it suffices to compute $D'(x)$.

Let:

$$D(x) = \sqrt{A^2x + s^2} \cdot \sqrt{B^2x + b^2}.$$

Using the product rule and chain rule:

$$D'(x) = \frac{A^2}{2\sqrt{A^2x + s^2}} \cdot \sqrt{B^2x + b^2} + \sqrt{A^2x + s^2} \cdot \frac{B^2}{2\sqrt{B^2x + b^2}}.$$

Therefore:

$$f'(x) = \frac{AB}{D(x)^2} \left[D(x) - \frac{x}{2} \left(\frac{A^2\sqrt{B^2x + b^2}}{\sqrt{A^2x + s^2}} + \frac{B^2\sqrt{A^2x + s^2}}{\sqrt{B^2x + b^2}} \right) \right].$$

This expression is complex but observe the following:

- For all $x > 0$, $f(x)$ is continuous and differentiable. - The numerator grows linearly in x , while the denominator grows sublinearly (square roots of linear functions).

Indeed, We now prove the expression in square brackets is positive. Consider the term:

$$uv - \frac{x}{2} \left(\frac{A^2v}{u} + \frac{B^2u}{v} \right).$$

where

$$u := \sqrt{A^2x + s^2}, \quad v := \sqrt{B^2x + b^2}, \quad D(x) := uv.$$

Multiply numerator and denominator appropriately to find:

$$uv - \frac{x}{2} \left(\frac{A^2v}{u} + \frac{B^2u}{v} \right) = \frac{A^2b^2x + B^2s^2x + 2b^2s^2}{2uv}.$$

Therefore, the full derivative becomes:

$$f'(x) = \frac{AB}{D(x)^2} \cdot \frac{A^2b^2x + B^2s^2x + 2b^2s^2}{2uv}.$$

Since $x > 0$ and all parameters are positive (variances and squared coefficients), and as long as $AB > 0$, this implies:

$$f'(x) > 0.$$

- Thus, for all $A, B \neq 0$, the *magnitude* of $f(x)$ increases with x .

Limiting behavior:

- As $x \rightarrow 0$, $f(x) \rightarrow 0$. - As $x \rightarrow \infty$,

$$f(x) \rightarrow \frac{ABx}{\sqrt{A^2x} \cdot \sqrt{B^2x}} = \frac{ABx}{|A||B|x} = \text{sign}(AB).$$

So:

$$\lim_{x \rightarrow \infty} |\rho_{SB}(\theta)| = 1.$$

Hence, the correlation increases monotonically in magnitude as inflation variance increases. When inflation volatility becomes extremely large, the stock and bond markets behave exactly in the same way in absolute terms and are indistinguishable from each other in terms of their absolute movements. Under a positive (negative) correlation regime, if one goes up, the other must always go up (down).

The function $|\rho_{SB}(\theta)|$ is strictly increasing in σ_π^2 , for fixed $\alpha_S(\theta), \alpha_B(\theta), \sigma_S, \sigma_B$.

Proposition 3: Transition Probability Amplifies Correlation

Let p_H denote the probability of being in a high-inflation regime, and let ρ_{SB}^L and ρ_{SB}^H denote the stock-bond correlations in the low- and high-inflation regimes, respectively. Suppose that the magnitude of correlation is greater in the high-inflation regime, i.e., $|\rho_{SB}^H| > |\rho_{SB}^L|$. Then the expected stock-bond correlation is increasing in p_H :

$$\frac{\partial \mathbb{E}[\rho_{SB}]}{\partial p_H} > 0. \quad (9)$$

Proof: Define the expected stock-bond correlation as the regime-weighted average:

$$\mathbb{E}[\rho_{SB}] = (1 - p_H)\rho_{SB}^L + p_H\rho_{SB}^H. \quad (10)$$

This is a convex combination of the two regime-specific correlations, where $p_H \in [0, 1]$ and $(1 - p_H)$ is the probability of the low-inflation regime.

Differentiating $\mathbb{E}[\rho_{SB}]$ with respect to p_H yields:

$$\frac{d\mathbb{E}[\rho_{SB}]}{dp_H} = \frac{d}{dp_H} [(1 - p_H)\rho_{SB}^L + p_H\rho_{SB}^H] \quad (11)$$

$$= -\rho_{SB}^L + \rho_{SB}^H \quad (12)$$

$$= \rho_{SB}^H - \rho_{SB}^L. \quad (13)$$

Thus, the sign of the derivative depends entirely on the relative magnitude of ρ_{SB}^H and ρ_{SB}^L .

Now consider two cases:

Case 1: Positive Correlation Regimes. Suppose both correlations are positive: $\rho_{SB}^H > \rho_{SB}^L > 0$. Then clearly:

$$\frac{d\mathbb{E}[\rho_{SB}]}{dp_H} = \rho_{SB}^H - \rho_{SB}^L > 0.$$

Case 2: Negative Correlation Regimes. Suppose both are negative and $|\rho_{SB}^H| > |\rho_{SB}^L|$. Then:

$$\rho_{SB}^H < \rho_{SB}^L < 0 \quad \Rightarrow \quad \rho_{SB}^H - \rho_{SB}^L < 0.$$

So the expected correlation becomes more negative as $p_H \uparrow$, which means:

$$|\mathbb{E}[\rho_{SB}]| \text{ increases with } p_H.$$

Thus, regardless of the sign of correlation in either regime, an increase in the probability of the high-inflation regime leads to a stronger comovement in magnitude:

$$\frac{d}{dp_H} |\mathbb{E}[\rho_{SB}]| > 0. \tag{14}$$

This completes the proof.

Proposition 4: Asymmetric Inflation Betas

As asymmetry $\Delta\alpha$ increases:

$$\frac{\partial}{\partial \Delta\alpha} (\rho_{SB}^H - \rho_{SB}^L) > 0 \tag{15}$$

As asymmetry in inflation beta increases across regimes, the gap between high and low regime stock-bond correlations widens. Formally, let $\Delta\alpha$ denote the inflation beta differential across regimes. Then:

$$\frac{\partial}{\partial \Delta\alpha} (\rho_{SB}^H - \rho_{SB}^L) > 0, \tag{16}$$

indicating that greater asymmetry in inflation sensitivity amplifies the regime gap in stock-bond correlation.

Proof. Recall that the instantaneous correlation between stock and bond returns in regime θ is:

$$\rho_{SB}(\theta) = \frac{\alpha_S(\theta)\alpha_B(\theta)\sigma_\pi^2}{\sqrt{\alpha_S^2(\theta)\sigma_\pi^2 + \sigma_S^2} \cdot \sqrt{\alpha_B^2(\theta)\sigma_\pi^2 + \sigma_B^2}}.$$

Let us define $\rho_{SB}^H := \rho_{SB}(H)$ and $\rho_{SB}^L := \rho_{SB}(L)$. We now examine:

$$\Delta\rho := \rho_{SB}^H - \rho_{SB}^L.$$

Suppose for simplicity that all coefficients $\alpha_S(\theta), \alpha_B(\theta) > 0$ and fix all volatility terms and σ_π^2 .

Then:

$$\begin{aligned}\rho_{SB}^H &= \frac{\alpha_S(H)\alpha_B(H)\sigma_\pi^2}{\sqrt{\alpha_S^2(H)\sigma_\pi^2 + \sigma_S^2} \cdot \sqrt{\alpha_B^2(H)\sigma_\pi^2 + \sigma_B^2}}, \\ \rho_{SB}^L &= \frac{\alpha_S(L)\alpha_B(L)\sigma_\pi^2}{\sqrt{\alpha_S^2(L)\sigma_\pi^2 + \sigma_S^2} \cdot \sqrt{\alpha_B^2(L)\sigma_\pi^2 + \sigma_B^2}}.\end{aligned}$$

As $\Delta\alpha$ increases—i.e., the inflation betas in the high-inflation regime rise further above those in the low-inflation regime—the numerator in ρ_{SB}^H increases multiplicatively while the denominator grows more slowly (due to sublinear square root growth). Meanwhile, ρ_{SB}^L remains fixed. Therefore:

$$\frac{\partial}{\partial \Delta\alpha}(\rho_{SB}^H - \rho_{SB}^L) > 0.$$

More formally, let the inflation betas in the high inflation regime be:

$$\alpha_S^H = \alpha + \Delta\alpha, \quad \alpha_B^H = \alpha + \Delta\alpha,$$

and in the low inflation regime:

$$\alpha_S^L = \alpha, \quad \alpha_B^L = \alpha.$$

Let the correlation in regime $\theta \in \{H, L\}$ be:

$$\rho_{SB}^\theta = \frac{\alpha_S^\theta \alpha_B^\theta \sigma_\pi^2}{\sqrt{(\alpha_S^\theta)^2 \sigma_\pi^2 + \sigma_S^2} \cdot \sqrt{(\alpha_B^\theta)^2 \sigma_\pi^2 + \sigma_B^2}}.$$

Denote $\rho_{SB}^H = \rho_H$ and $\rho_{SB}^L = \rho_L$, and define the correlation gap:

$$\Delta\rho := \rho_H - \rho_L.$$

Differentiating $\Delta\rho$ with respect to $\Delta\alpha$ yields:

$$\frac{\partial\Delta\rho}{\partial\Delta\alpha} = \frac{\sigma_\pi^2(\alpha + \Delta\alpha)}{(b^2 + \sigma_\pi^2(\alpha + \Delta\alpha)^2)^{3/2}(s^2 + \sigma_\pi^2(\alpha + \Delta\alpha)^2)^{3/2}} \cdot \left[\begin{aligned} &2(b^2 + \sigma_\pi^2(\alpha + \Delta\alpha)^2)(s^2 + \sigma_\pi^2(\alpha + \Delta\alpha)^2) \\ &- \sigma_\pi^2(\alpha + \Delta\alpha)^2(b^2 + \sigma_\pi^2(\alpha + \Delta\alpha)^2 + s^2 + \sigma_\pi^2(\alpha + \Delta\alpha)^2) \end{aligned} \right].$$

This expression is strictly positive for all $\Delta\alpha > 0$ and $\alpha > 0$, since the growth of the numerator (linear in $\Delta\alpha$) dominates the growth of the denominator (sublinear due to square roots), confirming the proposition.

Proposition 5: Convex Amplification via Sensitivity Updates

If the inflation sensitivities α_S^H and α_B^H increase with the high-inflation regime probability p_H , then the expected stock-bond correlation is convex in p_H . Formally:

$$\frac{\partial^2 \mathbb{E}[\rho_{SB}]}{\partial p_H^2} > 0. \quad (17)$$

Proof.

Let the stock-bond correlation in the high and low inflation regimes be ρ_H and ρ_L respectively, and define the expected correlation as:

$$\mathbb{E}[\rho_{SB}] = p_H \cdot \rho_H + (1 - p_H) \cdot \rho_L. \quad (18)$$

Then:

$$\frac{\partial \mathbb{E}[\rho_{SB}]}{\partial p_H} = \rho_H - \rho_L + p_H \cdot \frac{\partial \rho_H}{\partial p_H} - (1 - p_H) \cdot \frac{\partial \rho_L}{\partial p_H}. \quad (19)$$

Assume ρ_L is approximately constant (or weakly dependent on p_H), while ρ_H depends on p_H through its dependence on α_S^H and α_B^H . That is, $\alpha_S^H = \alpha_S^H(p_H)$ and $\alpha_B^H = \alpha_B^H(p_H)$ are increasing functions of p_H .

Using the chain rule:

$$\frac{\partial \rho_H}{\partial p_H} = \frac{\partial \rho_H}{\partial \alpha_S^H} \cdot \frac{d\alpha_S^H}{dp_H} + \frac{\partial \rho_H}{\partial \alpha_B^H} \cdot \frac{d\alpha_B^H}{dp_H}. \quad (20)$$

Assuming both $\frac{d\alpha_S^H}{dp_H} > 0$ and $\frac{d\alpha_B^H}{dp_H} > 0$, and that ρ_H is convex in both α_S^H and α_B^H (as seen

in Proposition 2), then the second derivative of the expected correlation is:

$$\begin{aligned}\frac{\partial^2 \mathbb{E}[\rho_{SB}]}{\partial p_H^2} &= \frac{\partial}{\partial p_H} \left(\rho_H - \rho_L + p_H \cdot \frac{\partial \rho_H}{\partial p_H} \right) \\ &= \frac{\partial \rho_H}{\partial p_H} + p_H \cdot \frac{\partial^2 \rho_H}{\partial p_H^2}.\end{aligned}$$

This is positive under our assumptions, completing the proof.

Lemma: Invariance of Correlation Under Different Returns Frequency. Correlation Preservation under GBM:

Let the stock price $S(t)$ and the bond price $B(t)$ follow correlated geometric Brownian motions,

$$\frac{dS(t)}{S(t)} = \mu_S dt + \sigma_S dW_S(t), \quad \frac{dB(t)}{B(t)} = \mu_B dt + \sigma_B dW_B(t),$$

with instantaneous correlation $\mathbb{E}[dW_S(t) dW_B(t)] = \rho dt$, $\rho \in [-1, 1]$. For any fixed horizon $T > 0$, define cumulative arithmetic returns

$$R^S(T) = \int_0^T \frac{dS(t)}{S(t)}, \quad R^B(T) = \int_0^T \frac{dB(t)}{B(t)}.$$

Then

$$\text{Corr}[R^S(T), R^B(T)] = \rho.$$

Hence, the sample correlation computed from discrete-time returns over length T is a consistent estimator of the instantaneous correlation under the GBM assumption.

Integrating the SDEs over $[0, T]$ yields

$$R^S(T) = \mu_S T + \sigma_S W_S(T), \quad R^B(T) = \mu_B T + \sigma_B W_B(T).$$

The deterministic drifts $\mu_i T$ do not affect second moments. Therefore

$$\text{Cov}[R^S, R^B] = \sigma_S \sigma_B \text{Cov}[W_S(T), W_B(T)] = \sigma_S \sigma_B \rho T,$$

while $\text{Var}[R^S(T)] = \sigma_S^2 T$ and $\text{Var}[R^B(T)] = \sigma_B^2 T$. Dividing covariance by the product of standard deviations,

$$\text{Corr}[R^S, R^B] = \frac{\sigma_S \sigma_B \rho T}{\sqrt{\sigma_S^2 T \sigma_B^2 T}} = \rho.$$

Thus, under GBM, instantaneous and horizon- T correlations coincide; thus, month- or quarter-end return correlations consistently estimate the model's instantaneous parameter ρ .